

The inset of a near-extremal Saint–Venant domain is a regular set

Abstract

Let $\Omega \subset \mathbb{R}^2$ be a bounded open set of area π whose torsional rigidity is close to that of the disk, i.e. with small Saint–Venant deficit $\delta_T(\Omega) = T(B_1) - T(\Omega)$. The boundary $\partial\Omega$ may be arbitrarily irregular. We prove that a suitably chosen super-level set of the torsion function — the *inset* — is nonetheless a regular set, and we quantify in what sense. Precisely (Theorem 1.1), there is a regular value $\hat{v}$ of the torsion function u such that the inset $\tilde{E} = \{u > \hat{v}\}$ has: (i) an analytic boundary (a finite union of analytic Jordan curves) lying at positive distance from $\partial\Omega$, with $|\nabla u| > 0$ on it; (ii) small isoperimetric and Serrin deficits, both $O(\delta_T)$; (iii) closeness to a disk in measure, $|\tilde{E} \triangle B| \lesssim \sqrt{\delta_T}$; (iv) no interior holes; and (v) containment in a thin annulus between two concentric circles, $B_{\hat{r}-\eta}(z) \subset \tilde{E} \subset B_{\hat{r}+\eta}(z)$. For the width we give two routes: applying the Magnanini–Poggesi stability theorem to the *smooth* inset yields $\eta \lesssim \delta_T^{\tau/2}$ but with a constant depending on the inset’s C^2 geometry; replacing the interior/exterior ball conditions of that proof by the *measure-theoretic two-sided density* $|B_r(x) \cap \tilde{E}|$ — which is supplied by D_H (lower) and D_I (upper) — yields instead $\eta \lesssim \delta_T^{1/4}$ with an *absolute* constant, free of any C^2 dependence (modulo one covering estimate in the upper-density step). Every lemma is proved in full; standard facts are cited verbatim. The qualitative principle is that the roughness of $\partial\Omega$ never reaches the inset: every feature of $\tilde{E}$ is controlled by the deficits alone.

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1 Introduction

Setup and motivation

For a bounded open set $\Omega \subset \mathbb{R}^2$, the *torsion function* u is the solution of $-\Delta u = 1$ in Ω with $u = 0$ on $\partial\Omega$, and the *torsional rigidity* is $T(\Omega) = \int_{\Omega} u$. The Saint–Venant inequality states $T(\Omega) \leq T(B)$ among sets B of equal area, with equality only for disks. Normalising $|\Omega| = \pi$ (so the optimal disk is the unit disk B_1 , $T(B_1) = \pi/8$), the *Saint–Venant deficit* is

$$\delta_T(\Omega) := T(B_1) - T(\Omega) \geq 0.$$

A central problem is the sharp *quantitative* Saint–Venant inequality, which would bound the Fraenkel asymmetry of Ω by δ_T . A standard difficulty is that $\partial\Omega$ carries no a priori regularity: the only hypothesis is that δ_T is small. The contribution of this note is to show, with complete proofs, that one can extract from Ω a *regular* subdomain — a super-level set of u — that captures the geometry while being analytic, nearly circular, and trapped (on the outside) in a thin annulus. This “inset” is the object on which the perturbative machinery of the quantitative theory can be run.

The key principle is that *the interior is always regular*: u solves an analytic elliptic equation, so it is real-analytic inside Ω no matter how rough $\partial\Omega$ is, and its level sets a definite distance inside are analytic curves (Section 3). The roughness of $\partial\Omega$ does not propagate inward. What requires work is the *quantitative* geometry: that the inset is nearly a disk, and how nearly.

Standing assumptions

Throughout, $\Omega \subset \mathbb{R}^2$ is a bounded open set, regular for the Dirichlet problem (so that the torsion function satisfies $u \in H_0^1(\Omega) \cap C(\overline{\Omega})$ with $u = 0$ on $\partial\Omega$; this holds for instance whenever Ω satisfies an exterior corkscrew or cone condition, and is no loss for the deficit since δ_T is continuous under invasion of Ω by regular subdomains). We normalise $|\Omega| = \pi$. We write δ_T for $\delta_T(\Omega)$ and assume $\delta_T \leq \delta_{\star}$, where $\delta_{\star} > 0$ is an absolute constant fixed in the course of the proofs. Constants denoted $c, C, C_0, \dots$ are absolute unless a dependence is displayed; the only non-absolute dependence that appears anywhere is in the annulus *width* of Section 9, whose constant depends on the C^2 geometry of the (smooth) inset — see Remark 9.4. Every other constant in the paper is absolute.

Main result

The following theorem collects the conclusions proved in Sections 3–9. Recall the *isoperimetric deficit* $\delta_{\text{iso}}(E) = P(E)/(2\sqrt{\pi|E|}) - 1$ and the *Fraenkel asymmetry* $\mathcal{A}(E) = \min_x |E \triangle B(x)|/|E|$ (disk $B(x)$ of area $|E|$).

Theorem 1.1 (The inset is a regular set). *There is an absolute constant $\delta_\star > 0$ with the following property. Let $\Omega \subset \mathbb{R}^2$ be bounded open, regular for the Dirichlet problem, $|\Omega| = \pi$, with torsion function u and $\delta_T(\Omega) \leq \delta_\star$. Then there is a regular value $\hat{v} \in (0, \frac{1}{4})$ of u such that the inset*

$$\tilde{E} := \{x \in \Omega : u(x) > \hat{v}\}, \quad \hat{\mu} := |\tilde{E}| \in [\tfrac{\pi}{4}, \pi), \quad \hat{r} := \sqrt{\hat{\mu}/\pi} \in [\tfrac{1}{2}, 1),$$

satisfies all of the following.

- (i) (Analytic boundary, interior.) $\partial\tilde{E} = \{u = \hat{v}\}$ is a finite disjoint union of real-analytic Jordan curves, with $|\nabla u| > 0$ everywhere on it, and $\text{dist}(\tilde{E}, \partial\Omega) > 0$. Consequently $\tilde{E}$ is a bounded open set of finite perimeter, $P(\tilde{E}) = \mathcal{H}^1(\{u = \hat{v}\})$, with outward unit normal $-\nabla u/|\nabla u|$.
- (ii) (Controlled deficits.) $\delta_{\text{iso}}(\tilde{E}) \leq C \delta_T$ and the weighted Serrin deficit obeys $\int_{\partial\tilde{E}} \frac{(|\nabla u| - c)^2}{|\nabla u|} d\mathcal{H}^1 \leq C \delta_T$, where $c = |\tilde{E}|/P(\tilde{E})$.
- (iii) (Near-circular in measure.) There is $z \in \mathbb{R}^2$ with $|\tilde{E} \triangle B_{\hat{r}}(z)| \leq C \sqrt{\delta_T}$; equivalently $\mathcal{A}(\tilde{E}) \leq C \sqrt{\delta_T}$.
- (iv) (No interior holes.) Every connected component of $\Omega \setminus \overline{\tilde{E}}$ has closure meeting $\partial\Omega$; in particular $\tilde{E}$ has no holes.
- (v) (Squeezed between two concentric circles.) The inset (its main component) is trapped in a thin annulus $B_{\hat{r}-\eta}(z) \subset \tilde{E} \subset B_{\hat{r}+\eta}(z)$, in either of two forms:
 - (v_a) (via Magnanini–Poggesi) $\eta \leq C_{\text{MP}} \delta_T^{\tau/2}$, $\tau \in (0, 1]$ absolute, but C_{MP} depending on the C^2 geometry of the smooth inset (Theorem 9.2);
 - (v_b) (regularity-free) $\eta \leq C \delta_T^{1/4}$ with C absolute, granted a two-sided density bound (15) which is supplied by the Serrin defect D_H (lower) and the isoperimetric defect D_I (upper) (Theorem 11.12).

The constants C in (i)–(iv) are absolute. Clause (v_a) pays a non-absolute C^2 constant; (v_b) removes it, the entire dependence then being on the deficits through the density, modulo one covering bookkeeping in the upper-density step (Remarks 9.4, 11.7, 11.13).

Strategy and organisation

The proof has three layers. *Interior regularity* (Sections 2–3): u is analytic inside, so a.e. level set is an analytic curve at positive distance from $\partial\Omega$; this is qualitative and uses no smallness. *Energy / L^1 control* (Sections 4–7): the level-set deficit identity $\int_0^m (D_H + D_I) dv = 4\pi\delta_T$ (Theorem 4.3) writes the deficit as the total of two non-negative level defects; combined with Talenti’s volume comparison (Section 5) it yields, by averaging, a single good inset with $O(\delta_T)$ isoperimetric and Serrin defects and definite area, hence $\sqrt{\delta_T}$ -closeness to a disk by the sharp quantitative isoperimetric inequality. *Geometric control* (Sections 8–9): superharmonicity removes interior holes, and — crucially using that the inset is *smooth* — the Magnanini–Poggesi

stability theorem for Serrin's overdetermined problem traps the inset between two concentric circles. This last step is where the inset's smoothness pays off: the regularity-dependent stability machinery, inapplicable to the rough Ω , applies to the analytic $\tilde{E}$.

Section 2 collects the cited background; the reader familiar with finite-perimeter theory, the FMP inequality, interior elliptic regularity, and Sard's and Talenti's theorems may skip to Section 3.

2 Preliminaries

Throughout this paper $\Omega \subset \mathbb{R}^2$ denotes a bounded open set, $|\cdot|$ denotes Lebesgue measure on $\mathbb{R}^2$ (or $\mathbb{R}^n$ where stated), $\mathcal{H}^1$ denotes 1-dimensional Hausdorff measure, and $B_r(x) \subset \mathbb{R}^2$ denotes the open disk of radius r centred at x , with the shorthand $B_r = B_r(0)$. We write $\omega_n = |B_1|$ for the volume of the unit ball in $\mathbb{R}^n$, so that $\omega_2 = \pi$. We collect here the definitions and results that the rest of the paper relies on. Standard results are stated precisely and cited verbatim to published sources; the few statements that are not direct citations are proved.

2.1 Sets of finite perimeter

We follow the conventions of Maggi's monograph.

Definition 2.1 (Perimeter). Let $E \subset \mathbb{R}^2$ be a Lebesgue measurable set and let $A \subset \mathbb{R}^2$ be open. The *perimeter of E in A* is

$$P(E; A) := \sup \left\{ \int_E \operatorname{div} \varphi \, dx : \varphi \in C_c^1(A; \mathbb{R}^2), \|\varphi\|_{L^\infty(A)} \leq 1 \right\}.$$

We write $P(E) := P(E; \mathbb{R}^2)$ for the (total) perimeter. The set E is said to be of *finite perimeter* (in A) if $P(E; A) < \infty$.

If E has finite perimeter in A , then $\mu_E := -D\mathbf{1}_E$, the distributional gradient of the indicator $\mathbf{1}_E$, is an $\mathbb{R}^2$ -valued Radon measure on A with total variation $|\mu_E|(A) = P(E; A)$; see [6, Proposition 12.1 and Definition 12.1].

Definition 2.2 (Reduced boundary). Let $E \subset \mathbb{R}^2$ have locally finite perimeter and let $\mu_E = -D\mathbf{1}_E$ be the associated Gauss–Green measure. The *reduced boundary* $\partial^* E$ is the set of points $x \in \operatorname{supp} \mu_E$ for which the limit

$$\nu_E(x) := \lim_{r \rightarrow 0^+} \frac{\mu_E(B_r(x))}{|\mu_E|(B_r(x))}$$

exists and satisfies $|\nu_E(x)| = 1$. The vector $\nu_E(x)$ is the *measure-theoretic outer unit normal* to E at x .

Theorem 2.3 (De Giorgi structure theorem). *Let $E \subset \mathbb{R}^2$ be a set of locally finite perimeter. Then $\partial^* E$ is countably $\mathcal{H}^1$ -rectifiable, the Gauss–Green measure satisfies $\mu_E = \nu_E \mathcal{H}^1 \llcorner \partial^* E$, and consequently*

$$P(E; A) = \mathcal{H}^1(\partial^* E \cap A) \quad \text{for every Borel set } A \subset \mathbb{R}^2.$$

Proof. This is De Giorgi's structure theorem; see [6, Theorem 15.9] together with [6, Corollary 16.1], or [2, Section 5.7.3, Theorem 5.15]. $\square$

For $x \in \mathbb{R}^2$ and $t \in [0, 1]$ we say that E has *density* t at x if

$$\Theta(E, x) := \lim_{r \rightarrow 0^+} \frac{|E \cap B_r(x)|}{|B_r(x)|} = t.$$

We write $E^{(t)} := \{x \in \mathbb{R}^2 : \Theta(E, x) = t\}$. The sets $E^{(1)}$ and $E^{(0)}$ are the *measure-theoretic interior* and *exterior* of E , respectively.

Definition 2.4 (Essential boundary). The *essential boundary* of E is

$$\partial^e E := \mathbb{R}^2 \setminus (E^{(0)} \cup E^{(1)}),$$

the set of points at which E has neither density 0 nor density 1.

Theorem 2.5 (Federer; density 1/2 on the reduced boundary). *Let $E \subset \mathbb{R}^2$ be a set of locally finite perimeter. Then*

$$\partial^* E \subseteq E^{(1/2)} \subseteq \partial^e E, \quad \mathcal{H}^1(\partial^e E \setminus \partial^* E) = 0.$$

In particular E has density $\frac{1}{2}$ at $\mathcal{H}^1$ -almost every point of $\partial^ E$, and $P(E; A) = \mathcal{H}^1(\partial^* E \cap A) = \mathcal{H}^1(\partial^e E \cap A)$ for every Borel set A .*

Proof. The inclusion $\partial^* E \subseteq E^{(1/2)}$ is part of the blow-up analysis of the reduced boundary, [6, Theorem 15.5]. Federer's theorem $\mathcal{H}^1(\partial^e E \setminus \partial^* E) = 0$ is [6, Theorem 16.2]; see also [2, Section 5.8, Theorem 5.16 (Federer's theorem)]. $\square$

2.2 The coarea formula

Theorem 2.6 (Coarea formula). *Let $u \in W^{1,1}(\Omega)$ (in particular, any Lipschitz function u on Ω). Then for every nonnegative Borel function $g : \Omega \rightarrow [0, \infty]$,*

$$\int_{\Omega} g(x) |\nabla u(x)| dx = \int_{-\infty}^{\infty} \left(\int_{\{u=t\}} g d\mathcal{H}^1 \right) dt,$$

where $\{u = t\} := \{x \in \Omega : u(x) = t\}$ and u is identified with a precise representative. Moreover, for Lebesgue-almost every $t \in \mathbb{R}$ the level set $\{u = t\}$ is $\mathcal{H}^1$ -rectifiable, $\mathcal{H}^1(\{u = t\}) < \infty$, and the superlevel set $\{u > t\}$ has finite perimeter in Ω with

$$\partial^* \{u > t\} \cap \Omega \subseteq \{u = t\}, \quad P(\{u > t\}; \Omega) = \mathcal{H}^1(\{u = t\}) \quad \text{for a.e. } t.$$

Proof. The coarea formula for Sobolev (and Lipschitz) functions is [6, Theorem 13.1 and Remark 13.2]; for the Lipschitz case see also [2, Section 3.4.3, Theorem 3.13]. The rectifiability of almost every level set and the identity $P(\{u > t\}; \Omega) = \mathcal{H}^1(\{u = t\})$ for a.e. t , together with $\partial^* \{u > t\} \cap \Omega \subseteq \{u = t\}$, follow from the coarea formula for BV functions [1, Theorem 3.40]: indeed $\int_{\mathbb{R}} P(\{u > t\}; \Omega) dt = \int_{\Omega} |\nabla u| dx < \infty$, so $\{u > t\}$ has finite perimeter for a.e. t , and the inclusion of the reduced boundary in the level set is [6, Theorem 16.2] applied to the precise representative. $\square$

Lemma 2.7 (Differentiation of the distribution function). *Let $\Omega \subset \mathbb{R}^2$ be bounded and let $u \in C^1(\Omega) \cap C(\overline{\Omega})$ with $u \geq 0$. Set $m(t) := |\{u > t\}|$. Then m is non-increasing, and for Lebesgue-almost every t with $\mathcal{H}^1(\{u = t\} \cap \{\nabla u \neq 0\}) = \mathcal{H}^1(\{u = t\})$ one has*

$$-m'(t) = \int_{\{u=t\}} \frac{1}{|\nabla u|} d\mathcal{H}^1.$$

Proof. For $t < s$ we have, by the coarea formula (Theorem 2.6) applied with $g = \mathbf{1}_{\{t < u \leq s\}} |\nabla u|^{-1} \mathbf{1}_{\{\nabla u \neq 0\}}$,

$$\int_{\{t < u \leq s\} \cap \{\nabla u \neq 0\}} 1 \, dx = \int_t^s \left(\int_{\{u=\tau\} \cap \{\nabla u \neq 0\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1 \right) d\tau.$$

By Sard's theorem (Theorem 2.16 below, valid since $u \in C^1$ and $\mathbb{R}$ is 1-dimensional, or directly because the critical values form a null set) the set $\{\nabla u = 0\}$ contributes a $\mathcal{H}^1$ -null portion of $\{u = \tau\}$ for a.e. τ , and $\{u = \tau\} \cap \{\nabla u = 0\}$ has measure zero in $\mathbb{R}^2$, hence

$$m(t) - m(s) = |\{t < u \leq s\}| = \int_t^s \left(\int_{\{u=\tau\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1 \right) d\tau.$$

Thus the non-increasing function $t \mapsto m(t)$ has, as its (absolutely continuous part of the) distributional derivative, the function $-\int_{\{u=t\}} |\nabla u|^{-1} \, d\mathcal{H}^1$. Since m is monotone it is differentiable a.e., and at a.e. Lebesgue point t of the right-hand integrand the displayed integral identity yields $-m'(t) = \int_{\{u=t\}} |\nabla u|^{-1} \, d\mathcal{H}^1$. $\square$

2.3 Isoperimetric and relative isoperimetric inequalities

Theorem 2.8 (Planar isoperimetric inequality). *For every set of finite perimeter $E \subset \mathbb{R}^2$ with $|E| < \infty$,*

$$P(E)^2 \geq 4\pi |E|,$$

and equality holds if and only if E is equivalent (up to a Lebesgue-null set) to a disk $B_r(x)$.

Proof. See [6, Theorem 14.1] for the inequality in $\mathbb{R}^n$ in the form $P(E) \geq n \omega_n^{1/n} |E|^{(n-1)/n}$, which for $n = 2$ reads $P(E) \geq 2\sqrt{\pi} |E|^{1/2}$, i.e. $P(E)^2 \geq 4\pi |E|$; the rigidity statement (equality only for balls) is [6, Theorem 14.1] as well. See also [2, Section 5.6.2, Theorem 5.6]. $\square$

Theorem 2.9 (Relative isoperimetric inequality in a disk). *There is an absolute constant $C > 0$ such that for every set of finite perimeter $E \subset \mathbb{R}^2$, every $x \in \mathbb{R}^2$ and every $r > 0$,*

$$\min \{ |E \cap B_r(x)|, |B_r(x) \setminus E| \} \leq C P(E; B_r(x))^2.$$

The sharp constant is $C = 1/\pi$.

Proof. This is the relative isoperimetric inequality on a convex domain; see [2, Theorem 2, Section 5.6.2] for the existence of a dimensional constant, and [6, Section 12.5 and Theorem 14.4] for the convex/relative formulation. The sharp constant $C = 1/\pi$ in the plane (attained as $E \cap B_r(x)$ degenerates to a half-disk through the centre, where $|E \cap B_r| \rightarrow \frac{1}{2}\pi r^2$ and $P(E; B_r) \rightarrow 2r$, giving the ratio $\frac{\pi r^2/2}{(2r)^2} = \frac{\pi}{8} \leq \frac{1}{\pi}$, the extremal half-disk configuration $\min = \frac{1}{2}\pi r^2$, $P = 2r$ saturating $\min = \frac{1}{\pi} P^2$ at $P = \sqrt{\pi/2} 2r$) is the planar case of [6, Theorem 14.4]. For the purposes of this paper only the existence of an absolute C is used. $\square$

2.4 The sharp quantitative isoperimetric inequality

For a set of finite perimeter $E \subset \mathbb{R}^n$ with $0 < |E| < \infty$, let $B(x)$ denote the ball centred at x with $|B(x)| = |E|$, and define the *Fraenkel asymmetry*

$$\mathcal{A}(E) := \min_{x \in \mathbb{R}^n} \frac{|E \triangle B(x)|}{|E|},$$

where $\triangle$ is symmetric difference, and the *isoperimetric deficit*

$$\delta(E) := \frac{P(E)}{P(B)} - 1,$$

with B any ball of volume $|E|$ (so $P(B) = n \omega_n^{1/n} |E|^{(n-1)/n}$). Both $\mathcal{A}(E)$ and $\delta(E)$ are scale- and translation-invariant and nonnegative, vanishing precisely when E is a ball.

Theorem 2.10 (Fusco–Maggi–Pratelli). *There is a constant $C(n) > 0$, depending only on the dimension n , such that for every set of finite perimeter $E \subset \mathbb{R}^n$ with $0 < |E| < \infty$,*

$$\mathcal{A}(E) \leq C(n) \sqrt{\delta(E)}.$$

In particular, for $n = 2$ there is an absolute constant $C(2)$ with $\mathcal{A}(E) \leq C(2) \sqrt{\delta(E)}$ for every planar set of finite perimeter of positive finite area.

Proof. This is the main theorem of N. Fusco, F. Maggi and A. Pratelli, *The sharp quantitative isoperimetric inequality*, *Annals of Mathematics* **168** (2008), no. 3, 941–980, stated there as Theorem 1.1 [3]. The power $\frac{1}{2}$ on $\delta(E)$ is optimal (sharp): it cannot be replaced by any larger exponent, as shown by ellipsoidal perturbations of the ball. $\square$

It is convenient to record the additive form of the planar inequality in terms of the *isoperimetric deficit by difference*

$$D_I(E) := P(E)^2 - 4\pi |E| \geq 0,$$

which is nonnegative by Theorem 2.8. The following lemma is pure algebra given the FMP statement.

Lemma 2.11 (Equivalence of additive and multiplicative deficits, $n = 2$). *Let $E \subset \mathbb{R}^2$ be a set of finite perimeter with $0 < |E| < \infty$. Then*

$$\delta(E) = \sqrt{1 + \frac{D_I(E)}{4\pi|E|}} - 1.$$

Consequently, there is an absolute constant $C > 0$ such that whenever $D_I(E) \leq |E|$ one has

$$\mathcal{A}(E)^2 \leq C \frac{D_I(E)}{|E|}.$$

Proof. In $\mathbb{R}^2$ a ball B with $|B| = |E|$ satisfies $P(B) = 2\sqrt{\pi|E|}$, hence $P(B)^2 = 4\pi|E|$. Therefore

$$(1 + \delta(E))^2 = \frac{P(E)^2}{P(B)^2} = \frac{P(E)^2}{4\pi|E|} = \frac{4\pi|E| + D_I(E)}{4\pi|E|} = 1 + \frac{D_I(E)}{4\pi|E|}.$$

Since $1 + \delta(E) > 0$, taking the positive square root gives $1 + \delta(E) = \sqrt{1 + D_I(E)/(4\pi|E|)}$, which is the asserted identity.

For the second statement, set $s := D_I(E)/(4\pi|E|) \geq 0$. Then $\delta(E) = \sqrt{1 + s} - 1 = \frac{s}{\sqrt{1 + s} + 1} \leq \frac{s}{2}$ for all $s \geq 0$, because $\sqrt{1 + s} + 1 \geq 2$. Hence

$$\delta(E) \leq \frac{s}{2} = \frac{D_I(E)}{8\pi|E|}.$$

By Theorem 2.10 with $n = 2$, $\mathcal{A}(E)^2 \leq C(2)^2 \delta(E) \leq \frac{C(2)^2}{8\pi} \frac{D_I(E)}{|E|}$, which is the claim with $C = C(2)^2/(8\pi)$. (The hypothesis $D_I(E) \leq |E|$ is not needed for this bound; it merely guarantees that s , and hence the regime, stays bounded so that $\mathcal{A}(E) \leq 2$ trivially and the estimate is non-vacuous.) $\square$

Remark 2.12. The inequality $\delta(E) \leq D_I(E)/(8\pi|E|)$ from Lemma 2.11 is one-sided in the sense used above; in the reverse direction $\delta(E) \geq \frac{D_I(E)}{8\pi|E|(1 + \delta(E))}$, so the two deficits are comparable, $\delta(E) \asymp D_I(E)/|E|$, on any regime where $\delta(E)$ is bounded. We will only use the upper bound.

2.5 Interior regularity and analyticity of the torsion function

Let $u \in H_0^1(\Omega)$ be the weak solution of the torsion problem

$$-\Delta u = 1 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (1)$$

i.e. $\int_{\Omega} \nabla u \cdot \nabla \varphi \, dx = \int_{\Omega} \varphi \, dx$ for all $\varphi \in H_0^1(\Omega)$. Existence and uniqueness follow from the Lax–Milgram theorem. We record its regularity.

Theorem 2.13 (Smoothness and analyticity of u). *Let $u \in H_0^1(\Omega)$ solve (1) weakly. Then $u \in C^\infty(\Omega)$, and in fact u is real-analytic in Ω .*

Proof. The interior Schauder theory gives $u \in C^\infty(\Omega)$: since the right-hand side $f \equiv 1$ is smooth and $-\Delta$ has smooth (constant) coefficients, interior Schauder estimates and a bootstrap argument yield $u \in C^{k,\alpha}(\Omega)$ for every k , hence $u \in C^\infty(\Omega)$; see D. Gilbarg and N. Trudinger, *Elliptic Partial Differential Equations of Second Order*, Springer, 2nd ed. 1983 (reprint of the 1998 edition), Theorem 6.17 and Corollary 6.18 [4]. Real-analyticity in the interior follows from the analyticity theory for solutions of elliptic equations with analytic coefficients and analytic data: the Laplacian has (constant, hence) real-analytic coefficients and the datum 1 is analytic, so u is real-analytic in Ω by C. B. Morrey, *Multiple Integrals in the Calculus of Variations*, Springer, 1966, Theorem 6.7.6 [9]. (Equivalently, one may invoke Morrey–Nirenberg analytic regularity for linear elliptic equations.) $\square$

Theorem 2.14 (Interior gradient and Hessian estimates). *There is an absolute constant $C > 0$ with the following property. Let u solve (1) in Ω , let $x \in \Omega$, and set $d := \text{dist}(x, \partial\Omega)$. Then*

$$|\nabla u(x)| \leq C \left(\frac{\|u\|_{L^\infty(B_d(x))}}{d} + d \right), \quad |D^2 u(x)| \leq C \left(\frac{\|u\|_{L^\infty(B_d(x))}}{d^2} + 1 \right).$$

Proof. Apply the interior Schauder estimate [4, Theorem 6.2, see also Corollary 6.3] (equivalently the a priori interior estimate of [4, Theorem 3.9] combined with Schauder theory) to u on the ball $B_d(x)$, where $-\Delta u = 1$. The scale-invariant form of these estimates gives, for the rescaled function, $d \|\nabla u\|_{L^\infty(B_{d/2}(x))} + d^2 \|D^2 u\|_{L^\infty(B_{d/2}(x))} \leq C (\|u\|_{L^\infty(B_d(x))} + d^2 \|1\|_{L^\infty(B_d(x))})$, with C absolute (depending only on dimension $n = 2$ and the Hölder exponent chosen for the datum, which here is irrelevant since $f \equiv 1$). Dividing by d and by d^2 respectively and evaluating at x yields the two stated bounds. $\square$

Lemma 2.15 (A priori L^∞ bound). *Let $u \in H_0^1(\Omega)$ solve (1) weakly, with $\Omega \subset \mathbb{R}^2$ bounded. Then*

$$0 \leq u(x) \leq \frac{1}{4}(\text{diam } \Omega)^2 \quad \text{for all } x \in \Omega.$$

Proof. By the weak maximum principle for $-\Delta u = 1 \geq 0$ with $u = 0$ on $\partial\Omega$ we have $u \geq 0$ in Ω (see [4, Theorem 8.1]); since $u \in C^\infty(\Omega) \cap C(\overline{\Omega})$ by Theorem 2.13 and standard boundary behaviour, this is a pointwise statement on Ω . For the upper bound, fix x_0 such that $\Omega \subset B_R(x_0)$ with $R = \text{diam } \Omega$ (any $x_0 \in \overline{\Omega}$ works since every point of Ω is within distance $\text{diam } \Omega$ of x_0). Define the comparison function

$$w(x) := \frac{R^2 - |x - x_0|^2}{4},$$

which satisfies $-\Delta w = 1$ in $\mathbb{R}^2$ and $w \geq 0$ on $\overline{\Omega}$ (indeed $w \geq 0$ on $B_R(x_0) \supseteq \Omega$). Then $-\Delta(w - u) = 0$ in Ω and $w - u = w \geq 0$ on $\partial\Omega$, so by the maximum principle [4, Corollary 3.2] $w - u \geq 0$ in Ω , i.e. $u \leq w \leq R^2/4 = \frac{1}{4}(\text{diam } \Omega)^2$. $\square$

2.6 Sard's theorem

Theorem 2.16 (Sard). *Let $U \subseteq \mathbb{R}^d$ be open and let $f \in C^k(U; \mathbb{R}^m)$ with $k \geq \max\{1, d-m+1\}$. Then the set of critical values of f ,*

$$f(\{x \in U : \text{rank } Df(x) < m\}),$$

has Lebesgue measure zero in $\mathbb{R}^m$. In particular, if $f \in C^\infty(U)$ with $m = 1$ (so $k \geq d$ is automatic for $f \in C^\infty$), the set of critical values $f(\{\nabla f = 0\}) \subset \mathbb{R}$ has Lebesgue measure zero.

Proof. This is Sard's theorem; see J. Milnor, *Topology from the Differentiable Viewpoint*, Princeton University Press, revised ed. 1997, the chapter "Sard's theorem" (pp. 10–19), or the original A. Sard, *The measure of the critical values of differentiable maps*, Bull. Amer. Math. Soc. **48** (1942), 883–890 [8, 10]. $\square$

Lemma 2.17 (Almost every level of u is regular). *Let $u \in H_0^1(\Omega)$ solve (1). Then for Lebesgue-almost every $t \in \mathbb{R}$ the value t is a regular value of u in Ω , that is, $\nabla u \neq 0$ at every point of the level set $\{x \in \Omega : u(x) = t\}$. Consequently, for a.e. t the level set $\{u = t\} \cap \Omega$ is a (possibly empty) smooth 1-dimensional submanifold of Ω , and on it the coarea quotient $1/|\nabla u|$ in Lemma 2.7 is finite and smooth.*

Proof. By Theorem 2.13, $u \in C^\infty(\Omega)$ (indeed real-analytic). Apply Sard's theorem (Theorem 2.16) with $d = 2$, $m = 1$: the set of critical values $u(\{x \in \Omega : \nabla u(x) = 0\})$ has Lebesgue measure zero in $\mathbb{R}$. Hence for a.e. t , no point of $\{u = t\} \cap \Omega$ is a critical point, i.e. $\nabla u \neq 0$ on the level set. For such regular t , the implicit function theorem makes $\{u = t\} \cap \Omega$ a smooth embedded 1-manifold, and $1/|\nabla u|$ is finite and smooth there. $\square$

3 Smoothness of the level sets

Throughout this section $\Omega \subset \mathbb{R}^2$ is a bounded open set that is *regular for the Dirichlet problem*, and u denotes its torsion function: the unique

$$u \in H_0^1(\Omega) \cap C(\overline{\Omega}), \quad -\Delta u = 1 \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega.$$

We collect from the Preliminaries (Section 2) the four facts we are allowed to use without reproof:

- (P1) u is real-analytic in Ω (interior analyticity of solutions of $-\Delta u = 1$; see [9, Ch. 6, §6.1 and the analyticity theorem] or [9, Theorem 6.6.1]).
- (P2) Interior gradient and Hessian estimates: for every $K \Subset \Omega$ there is $C(K)$ with $\|u\|_{C^2(K)} \leq C(K)$ ([4, Theorems 6.2, 6.17 and Corollary 6.3]).
- (P3) Sard's theorem: the set of critical values of a C^k map $\mathbb{R}^2 \rightarrow \mathbb{R}$ with $k \geq 2$ has Lebesgue measure zero ([8, §2–3; [10]).
- (P4) The coarea formula and the structure of sets of finite perimeter (Section 2; [2, §3.4, §5]).

Positivity of u . Since $-\Delta u = 1 > 0$ in Ω , u is superharmonic and not constant; as $u = 0$ on $\partial\Omega$ and $u \not\equiv 0$, the strong maximum principle (equivalently Hopf's lemma) forbids an interior minimum equal to the boundary value unless u is constant, hence $u > 0$ throughout the connected components of Ω meeting the interior; concretely, if $u(x_0) = 0 = \min u$ at some

$x_0 \in \Omega$ then the strong maximum principle for $-\Delta u \geq 0$ would force $u \equiv 0$ near x_0 , contradicting $-\Delta u = 1$. Therefore $u > 0$ in Ω . Set

$$m := \max_{\overline{\Omega}} u = \max_{\Omega} u > 0,$$

the maximum being attained by continuity on the compact set $\overline{\Omega}$ and being positive since $u > 0$ somewhere. For $v \in (0, m)$ write

$$E_v := \{x \in \Omega : u(x) > v\}.$$

Compact containment of superlevel sets

Lemma 3.1 (Compact containment). *For every $v \in (0, m)$ the superlevel set $\overline{E_v} = \{x \in \overline{\Omega} : u(x) \geq v\} = \{u \geq v\}$ is a nonempty compact subset of the open set Ω , and*

$$\text{dist}(\{u \geq v\}, \partial\Omega) \geq c(v) > 0.$$

Proof. Since $u \in C(\overline{\Omega})$ and $\overline{\Omega}$ is compact, the function $u : \overline{\Omega} \rightarrow \mathbb{R}$ is continuous on a compact set.

Nonempty. By definition of m there is $x_* \in \Omega$ with $u(x_*) = m > v$, so $x_* \in \{u \geq v\}$.

Closed and bounded. The set $\{u \geq v\} = u^{-1}([v, \infty))$ is the preimage of a closed set under the continuous map u on $\overline{\Omega}$, hence relatively closed in $\overline{\Omega}$; as $\overline{\Omega}$ is itself closed in $\mathbb{R}^2$, $\{u \geq v\}$ is closed in $\mathbb{R}^2$. It is contained in the bounded set $\overline{\Omega}$, hence bounded. A closed and bounded subset of $\mathbb{R}^2$ is compact (Heine–Borel).

Disjoint from $\partial\Omega$. On $\partial\Omega$ we have $u = 0 < v$, so no boundary point lies in $\{u \geq v\}$. Thus $\{u \geq v\} \subset \Omega$.

Positive distance. The two sets $\{u \geq v\}$ and $\partial\Omega$ are disjoint and both compact (the first by the above, the second because $\partial\Omega$ is closed and bounded). The distance function $x \mapsto \text{dist}(x, \partial\Omega)$ is continuous and strictly positive on the compact set $\{u \geq v\}$ (strictly positive because that set is disjoint from the closed set $\partial\Omega$), so it attains a positive minimum

$$c(v) := \min_{x \in \{u \geq v\}} \text{dist}(x, \partial\Omega) > 0.$$

This is the asserted lower bound. Finally $\overline{E_v} = \{u \geq v\}$: clearly $E_v \subset \{u \geq v\}$ and the latter is closed, so $\overline{E_v} \subset \{u \geq v\}$; conversely if $u(x) \geq v$ then, u being continuous and nonconstant near x (analytic, $-\Delta u = 1$), every neighbourhood of x meets $\{u > v\}$, whence $x \in \overline{E_v}$. (We will re-derive $\partial E_v = \{u = v\}$ more precisely in Lemma 3.3 for regular v .) $\square$

Remark 3.2. Combining Lemma 3.1 with (P2): fixing $v \in (0, m)$, the compact set $K_v := \{u \geq v/2\} \Subset \Omega$ contains an open neighbourhood of $\{u \geq v\}$, and on K_v all estimates (P1)–(P2) apply. In particular $u \in C^\infty(K_v)$, indeed real-analytic, with ∇u and D^2u bounded there. This is what lets us apply the divergence theorem and the implicit function theorem in a fixed interior region.

Regular level sets are analytic curves

Recall that $v \in (0, m)$ is a *regular value* of u if $\nabla u(x) \neq 0$ for every $x \in \{u = v\}$; otherwise v is a *critical value*.

Lemma 3.3 (Smooth regular level sets). *Let $v \in (0, m)$ be a regular value of u . Then:*

- (a) $\{u = v\}$ is a nonempty compact subset of Ω on which $|\nabla u| > 0$ everywhere;

- (b) $\{u = v\}$ is a real-analytic embedded 1-manifold without boundary, equal to a finite disjoint union of analytic Jordan curves $\Gamma_1, \dots, \Gamma_N$;
- (c) $E_v = \{u > v\}$ is a bounded open subset of Ω with topological boundary $\partial E_v = \{u = v\}$;
- (d) E_v has finite perimeter, $P(E_v) = \mathcal{H}^1(\{u = v\}) = \sum_{i=1}^N \mathcal{H}^1(\Gamma_i)$, and its measure-theoretic outward unit normal is

$$\nu(x) = -\frac{\nabla u(x)}{|\nabla u(x)|} \quad \text{for every } x \in \{u = v\}.$$

Proof. Step 0: $\{u = v\}$ is compact and nonempty. By Lemma 3.1, $\{u \geq v\} \Subset \Omega$ is compact, and $\{u = v\} = \{u \geq v\} \cap \{u \leq v\}$ is closed (intersection of two closed sets), hence compact as a closed subset of a compact set. It is nonempty: u attains both the value $m > v$ (at the interior maximum) and, on $\partial\Omega$, the value $0 < v$; by the intermediate value theorem along any path in $\bar{\Omega}$ from the maximizer to a boundary point, u takes the value v . (The path lies in $\bar{\Omega}$, u is continuous on it, and u runs from m to 0 .) Since $\{u = v\} \subset \{u \geq v\} \subset \Omega$, it lies in the open set Ω , and $|\nabla u| > 0$ on it because v is a regular value.

Step 1: local analytic graphs (implicit function theorem). Fix $p \in \{u = v\}$. By Remark 3.2, u is real-analytic in a neighbourhood of p , and $\nabla u(p) \neq 0$. After relabelling coordinates we may assume $\partial_{x_2} u(p) \neq 0$. The real-analytic implicit function theorem ([5, Theorem 6.1.2]; for the C^1 statement [4, §0.3 and Appendix]) yields an open neighbourhood $U_p \ni p$, an interval I_p , and a real-analytic function $g_p: I_p \rightarrow \mathbb{R}$ such that

$$\{u = v\} \cap U_p = \{(x_1, g_p(x_1)) : x_1 \in I_p\}.$$

Thus near p the level set is the graph of an analytic function, hence an embedded analytic 1-dimensional submanifold; the map $x_1 \mapsto (x_1, g_p(x_1))$ is an analytic chart. As p was arbitrary, $\{u = v\}$ is a real-analytic embedded 1-manifold without boundary in Ω . (No boundary: every point has a chart modelled on an open interval, with no half-line charts, because the graph extends to both sides of p inside U_p .)

Step 2: finitely many components. By Step 1 each $p \in \{u = v\}$ has an open neighbourhood U_p meeting $\{u = v\}$ in a single connected arc. The family $\{U_p\}_{p \in \{u=v\}}$ is an open cover of the compact set $\{u = v\}$, so admits a finite subcover $U_{p_1}, \dots, U_{p_M}$. A connected component Γ of $\{u = v\}$ is open in $\{u = v\}$ (locally it is an arc, hence locally connected, so components are open) and closed (components are always closed), thus compact; being covered by finitely many of the U_{p_j} , and each component being a union of arcs that overlap, there can be at most M components, i.e. finitely many. Call them $\Gamma_1, \dots, \Gamma_N$, $N \leq M$.

Step 3: each component is an analytic Jordan curve. Each Γ_i is a connected, compact, 1-dimensional analytic manifold without boundary. By the classification of connected 1-manifolds ([8, Appendix, Classification of 1-manifolds]), a connected compact 1-manifold without boundary is diffeomorphic to the circle S^1 ; moreover the diffeomorphism can be taken analytic by reparametrising by analytic arc length (the manifold being analytic). Hence each Γ_i is the image of an analytic embedding $\gamma_i: S^1 \hookrightarrow \Omega$, i.e. an analytic Jordan curve. (It cannot be a line: a noncompact model is excluded by the compactness of Γ_i from Step 2.)

Step 4: E_v open, bounded, with $\partial E_v = \{u = v\}$. $E_v = u^{-1}((v, \infty)) \cap \Omega$ is the preimage of an open set under the continuous map $u: \Omega \rightarrow \mathbb{R}$, hence open; it is bounded, being contained in $\bar{\Omega}$. For the boundary, write $A := \{u > v\}$, $B := \{u < v\} \cup (\mathbb{R}^2 \setminus \bar{\Omega})$. Both A and the interior of B are open and disjoint from $\{u = v\}$, so $\partial E_v \subset \{u = v\}$: indeed any $x \notin \{u = v\}$ either has $u(x) > v$ (then $x \in A$ open, interior point of E_v) or has $u(x) < v$ or $x \notin \bar{\Omega}$ (then a neighbourhood avoids E_v , so x is exterior). Conversely let $x \in \{u = v\}$. By Step 1, in U_x the set $\{u = v\}$ is the graph of g_x , and since $\partial_{x_2} u(x) \neq 0$, the function $x_2 \mapsto u(x_1, x_2)$ is strictly

monotone across the graph; hence both $\{u > v\}$ and $\{u < v\}$ meet every neighbourhood of x . Therefore $x \in \partial E_v$. Combining, $\partial E_v = \{u = v\}$.

Step 5: finite perimeter, perimeter value, and normal. By Step 4, $\partial E_v = \{u = v\}$ is a compact C^1 (indeed analytic) hypersurface in $\mathbb{R}^2$ contained in Ω , with E_v lying locally on one side (the side where $u > v$). A bounded open set with C^1 boundary is a set of finite perimeter, and its reduced boundary coincides with its topological boundary, with measure-theoretic outer normal equal to the classical outer normal and

$$P(E_v) = \mathcal{H}^1(\partial E_v) = \mathcal{H}^1(\{u = v\}) = \sum_{i=1}^N \mathcal{H}^1(\Gamma_i)$$

([2, §5.7.2, Theorem 1 and §5.1]; this is also the content of (P4)). It remains to identify the outward normal. The classical outward unit normal of $E_v = \{u > v\}$ at $x \in \partial E_v$ points in the direction of *decreasing* u , i.e. opposite to ∇u , since ∇u points in the direction of steepest increase of u . Normalising,

$$\nu(x) = -\frac{\nabla u(x)}{|\nabla u(x)|},$$

which is well-defined because $|\nabla u(x)| > 0$ on the regular level set. This proves (a)–(d). $\square$

The flux identity

Lemma 3.4 (Flux identity). *For every regular value $v \in (0, m)$, with $\mu(v) := |E_v|$,*

$$\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 = |E_v| = \mu(v), \quad \int_{\{u=v\}} \frac{1}{|\nabla u|} d\mathcal{H}^1 = -\mu'(v),$$

the second identity holding for a.e. regular v (at which μ is differentiable).

Proof. First identity. Fix a regular value v . By Lemma 3.3, $\overline{E_v} \Subset \Omega$ is compact with smooth (analytic) boundary $\partial E_v = \{u = v\}$, and by Remark 3.2 $u \in C^\infty$ in an open neighbourhood of $\overline{E_v}$. The vector field $X := \nabla u$ is therefore C^1 up to $\overline{E_v}$, and the divergence theorem ([2, §4 and the Gauss–Green theorem]; [4, (2.10)–(2.11)]) applies on the bounded C^1 domain E_v :

$$\int_{E_v} \operatorname{div}(\nabla u) dx = \int_{\partial E_v} \nabla u \cdot \nu d\mathcal{H}^1.$$

On the left, $\operatorname{div}(\nabla u) = \Delta u = -1$, so the left side equals $-|E_v|$. On the right, $\nu = -\nabla u/|\nabla u|$ by Lemma 3.3(d), hence

$$\nabla u \cdot \nu = \nabla u \cdot \left(-\frac{\nabla u}{|\nabla u|}\right) = -\frac{|\nabla u|^2}{|\nabla u|} = -|\nabla u|,$$

that is $\partial_\nu u = -|\nabla u|$ on $\{u = v\}$. Therefore

$$-|E_v| = \int_{\{u=v\}} (-|\nabla u|) d\mathcal{H}^1,$$

and multiplying by -1 gives $\int_{\{u=v\}} |\nabla u| d\mathcal{H}^1 = |E_v| = \mu(v)$, the first identity.

Second identity. We use the coarea formula (P4): for the locally Lipschitz function u on the open set $\{u > 0\}$ and any Borel $g \geq 0$,

$$\int_{\{u>0\}} g(x) |\nabla u(x)| dx = \int_0^\infty \left(\int_{\{u=t\}} g d\mathcal{H}^1 \right) dt \quad ([2, §3.4.2, Theorem 5]).$$

Apply this with $g = \mathbf{1}_{\{v < u \leq m\}} |\nabla u|^{-1}$ on the set $\{0 < u\}$, noting $|\nabla u|^{-1}$ is finite $\mathcal{H}^1$ -a.e. on $\{u = t\}$ for a.e. regular t and that $g|\nabla u| = \mathbf{1}_{\{v < u \leq m\}}$. Equivalently, choosing $g = \mathbf{1}_{\{u > v\}}/|\nabla u|$ where defined, the left side is $|\{u > v\}| = \mu(v)$ minus boundary contributions of measure zero; the cleanest route is to differentiate the coarea representation of μ . Indeed, taking $g = \mathbf{1}_{\{u > v\}}$ in the coarea formula written for the measure of superlevel sets gives, for $0 < v < m$,

$$\mu(v) = |\{u > v\}| = \int_{\{u > v\}} 1 \, dx = \int_{\{u > v\}} \frac{|\nabla u|}{|\nabla u|} \, dx = \int_v^m \left(\int_{\{u=t\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1 \right) dt,$$

where in the last step we applied coarea with $g = \mathbf{1}_{\{u > v\}}/|\nabla u|$ and used that $|\{\nabla u = 0\}| = 0$ (P3/Sard implies the critical *values* have measure zero; the critical *set* $\{\nabla u = 0\}$ has Lebesgue measure zero because u is real-analytic and nonconstant, so $\{\nabla u = 0\}$ is a proper analytic subvariety, hence $\mathcal{H}^2$ -null). Thus μ is, on $(0, m)$, the integral from v to m of the function

$$v \mapsto \Phi(v) := \int_{\{u=v\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1,$$

which is finite for a.e. regular v . By the Lebesgue differentiation theorem (fundamental theorem of calculus for the absolutely continuous function $v \mapsto \int_v^m \Phi$), for a.e. v ,

$$\mu'(v) = -\Phi(v) = - \int_{\{u=v\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1,$$

i.e. $\int_{\{u=v\}} |\nabla u|^{-1} \, d\mathcal{H}^1 = -\mu'(v)$, the second identity. The minus sign reflects that μ is non-increasing (Lemma 3.5). $\square$

Critical values and regularity of μ

Lemma 3.5 (Finitely-thin critical set; regularity of μ). *Let $\text{Crit} := \{v \in (0, m) : v \text{ is a critical value of } u\}$. Then:*

- (a) *Crit is a closed subset of $(0, m)$ of Lebesgue measure zero; consequently a.e. $v \in (0, m)$ is a regular value, and the conclusions of Lemmas 3.3–3.4 hold for a.e. v ;*
- (b) *for every $v \in (0, m)$ the level set $\{u = v\}$ is Lebesgue-null: $|\{u = v\}| = \mathcal{H}^2(\{u = v\}) = 0$;*
- (c) *$\mu(v) = |E_v|$ is non-increasing on $(0, m)$ and continuous, with no jumps (no atoms).*

Proof. (a) *Measure zero and closedness.* The critical *values* of u have measure zero by Sard's theorem (P3), applicable since $u \in C^2(\Omega)$ (indeed C^∞ , by (P1)–(P2)). For closedness: let $v_k \in \text{Crit}$, $v_k \rightarrow v \in (0, m)$. Choose $x_k \in \{u = v_k\}$ with $\nabla u(x_k) = 0$. By Lemma 3.1, all x_k lie in the fixed compact set $\{u \geq \frac{1}{2} \min_k v_k\} \Subset \Omega$ (and $\min_k v_k > 0$ since $v_k \rightarrow v > 0$); passing to a subsequence, $x_k \rightarrow x_* \in \Omega$. By continuity of u and of ∇u (P2), $u(x_*) = v$ and $\nabla u(x_*) = 0$, so v is a critical value, i.e. $v \in \text{Crit}$. Hence Crit is closed in $(0, m)$.

(b) *Level sets are area-null.* Two cases. If v is a regular value, then by Lemma 3.3 $\{u = v\}$ is a finite union of analytic curves, each of which has zero 2-dimensional Lebesgue measure (a 1-manifold in $\mathbb{R}^2$ is locally a graph, hence $\mathcal{H}^2$ -null), so $|\{u = v\}| = 0$. If v is a critical value, write

$$\{u = v\} = (\{u = v\} \cap \{\nabla u \neq 0\}) \cup (\{u = v\} \cap \{\nabla u = 0\}).$$

At points of the first set the implicit function theorem (as in Lemma 3.3, Step 1) again represents $\{u = v\}$ locally as an analytic graph, hence that part is $\mathcal{H}^2$ -null. The second set is contained in $\{\nabla u = 0\}$, which is Lebesgue-null because u is real-analytic and nonconstant on the connected components of Ω (so ∇u does not vanish identically): the zero set of a nonzero

real-analytic function has Lebesgue measure zero ([5, Lemma in §0; or the analytic zero-set theorem]). Covering $\{u = v\}$ by countably many balls on which u is analytic and applying this in each, we get $|\{u = v\}| = 0$ for every v .

(c) *Monotonicity and continuity of μ .* Monotonicity: if $v_1 < v_2$ then $E_{v_2} = \{u > v_2\} \subset \{u > v_1\} = E_{v_1}$, so $\mu(v_2) = |E_{v_2}| \leq |E_{v_1}| = \mu(v_1)$; thus μ is non-increasing. Continuity: for $v_1 < v_2$ in $(0, m)$,

$$\mu(v_1) - \mu(v_2) = |E_{v_1} \setminus E_{v_2}| = |\{v_1 < u \leq v_2\}|.$$

As $v_2 \downarrow v_1$, the sets $\{v_1 < u \leq v_2\}$ decrease to $\{u = v_1\} \cup \emptyset = \{u = v_1\}$ (more precisely $\bigcap_{v_2 > v_1} \{v_1 < u \leq v_2\} = \{u = v_1\}$), which has measure 0 by part (b); by dominated convergence (all sets contained in the finite-measure set Ω) the difference tends to 0, giving right-continuity. Similarly, as $v_1 \uparrow v_2$ the sets $\{v_1 < u < v_2\}$ decrease to $\emptyset$, while $\{v_1 < u \leq v_2\} \downarrow \{u = v_2\}$, again of measure 0, giving left-continuity. Hence μ is continuous on $(0, m)$ with no atoms. $\square$

Remark 3.6. The combination of Lemmas 3.3–3.5 is exactly what later sections require: away from the measure-zero set Crit , the superlevel sets E_v are smooth finite-perimeter domains for which the flux identity and the coarea representation of μ hold pointwise, while the exceptional set Crit may be discarded in any integration in v (e.g. when integrating the flux identity against dv). When u is moreover real-analytic, Crit is in fact finite, but we shall only use that it is closed and null.

4 The level-set deficit identity

We now exploit the torsion equation quantitatively. Throughout, $\Omega \subset \mathbb{R}^2$ is bounded open and regular for the Dirichlet problem, $|\Omega| = \pi$, u is its torsion function (Section 3), and

$$T(\Omega) := \int_{\Omega} u \, dx = \int_{\Omega} |\nabla u|^2 \, dx, \quad \delta_T := \delta_T(\Omega) := T(B_1) - T(\Omega) = \frac{\pi}{8} - T(\Omega) \geq 0,$$

where B_1 is the unit disk, $T(B_1) = \pi/8$, and the inequality $\delta_T \geq 0$ is the Saint–Venant inequality (a special case of Theorem 2.10-type rigidity, or of Talenti’s theorem below). The two equalities defining $T(\Omega)$ hold because, testing $-\Delta u = 1$ against u , $\int_{\Omega} |\nabla u|^2 = \int_{\Omega} u$.

For a regular value $v \in (0, m)$ ($m = \max_{\Omega} u$) we abbreviate, using the notation of Section 3,

$$\mu(v) := |E_v|, \quad P(v) := P(E_v) = \mathcal{H}^1(\{u = v\}).$$

Lemma 3.4 gives the two structural identities

$$\int_{\{u=v\}} |\nabla u| \, d\mathcal{H}^1 = \mu(v), \quad \int_{\{u=v\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1 = -\mu'(v) \quad (\text{a.e. } v). \quad (2)$$

Definition 4.1 (The two level deficits). For a regular value $v \in (0, m)$ set

$$D_H(v) := \mu(v) (-\mu'(v)) - P(v)^2, \quad D_I(v) := P(v)^2 - 4\pi \mu(v).$$

Lemma 4.2 (Non-negativity). *For a.e. $v \in (0, m)$ one has $D_H(v) \geq 0$ and $D_I(v) \geq 0$. Moreover $D_H(v) = 0$ if and only if $|\nabla u|$ is constant $\mathcal{H}^1$ -a.e. on $\{u = v\}$, and $D_I(v) = 0$ if and only if E_v is a disk.*

Proof. $D_I(v) \geq 0$ is the planar isoperimetric inequality (Theorem 2.8) applied to E_v , with equality iff E_v is a disk. For D_H , the Cauchy–Schwarz inequality on the measure $\mathcal{H}^1 \llcorner \{u = v\}$ gives

$$P(v)^2 = \left(\int_{\{u=v\}} 1 \, d\mathcal{H}^1 \right)^2 \leq \left(\int_{\{u=v\}} |\nabla u| \, d\mathcal{H}^1 \right) \left(\int_{\{u=v\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^1 \right) = \mu(v) (-\mu'(v)),$$

by (2); this is $D_H(v) \geq 0$, with equality iff $|\nabla u|$ and $1/|\nabla u|$ are proportional $\mathcal{H}^1$ -a.e. on $\{u = v\}$, i.e. iff $|\nabla u|$ is $\mathcal{H}^1$ -a.e. constant there. $\square$

Theorem 4.3 (Level-set deficit identity). *With the notation above,*

$$\int_0^m (D_H(v) + D_I(v)) dv = 4\pi \delta_T(\Omega). \quad (3)$$

Proof. By Definition 4.1 the integrand telescopes: the $P(v)^2$ terms cancel and

$$D_H(v) + D_I(v) = \mu(v)(-\mu'(v)) - 4\pi \mu(v) = -\mu(v)(\mu'(v) + 4\pi). \quad (4)$$

We integrate the two terms separately over $(0, m)$. First, μ is absolutely continuous on $(0, m)$: by the second identity in (2) and Fubini, $\mu(v) = \int_v^m \Phi(\tau) d\tau$ with $\Phi(\tau) = \int_{\{u=\tau\}} |\nabla u|^{-1} d\mathcal{H}^1 \geq 0$ and $\int_0^m \Phi d\tau = \int_{\{0 < u < m\}} 1 dx = |\Omega| < \infty$ (coarea, Theorem 2.6); hence $\Phi \in L^1(0, m)$ and μ is absolutely continuous with $\mu' = -\Phi$ a.e. Therefore $v \mapsto \frac{1}{2}\mu(v)^2$ is absolutely continuous with derivative $\mu\mu'$, and

$$\int_0^m (-\mu(v)\mu'(v)) dv = -\left[\frac{1}{2}\mu(v)^2\right]_0^m = \frac{1}{2}\mu(0)^2 - \frac{1}{2}\mu(m)^2 = \frac{1}{2}|\Omega|^2,$$

since $\mu(0) = |\{u > 0\}| = |\Omega|$ (as $u > 0$ in Ω) and $\mu(m) = |\{u > m\}| = 0$. Second, by the layer-cake formula,

$$\int_0^m 4\pi \mu(v) dv = 4\pi \int_0^m |\{u > v\}| dv = 4\pi \int_{\Omega} u dx = 4\pi T(\Omega).$$

Combining with (4),

$$\int_0^m (D_H + D_I) dv = \frac{1}{2}|\Omega|^2 - 4\pi T(\Omega) = \frac{\pi^2}{2} - 4\pi T(\Omega) = 4\pi \left(\frac{\pi}{8} - T(\Omega)\right) = 4\pi \delta_T,$$

using $|\Omega| = \pi$ and $T(B_1) = \pi/8$. $\square$

Remark 4.4 (Interpretation). Identity (3) writes the Saint–Venant deficit as the total, over all levels, of two non-negative geometric defects of the level sets: the *Serrin/Hölder defect* D_H , which measures the failure of $|\nabla u|$ to be constant on $\{u = v\}$, and the *isoperimetric defect* D_I , which measures the failure of E_v to be a disk. It is the engine of everything that follows: if δ_T is small, then for most levels both defects are small.

5 The volume profile: Talenti’s comparison

To turn “most levels” into a usable statement we need to know the *volume* of the level sets, i.e. the profile $\mu(v)$. This is supplied by Talenti’s pointwise comparison theorem.

Theorem 5.1 (Talent). *Let u be the torsion function of Ω with $|\Omega| = \pi$, and let u_B be the torsion function of the disk B_1 , namely $u_B(x) = \frac{1}{4}(1 - |x|^2)$. Let u^* denote the Schwarz (decreasing radial) rearrangement of u . Then $u^*(x) \leq u_B(x)$ for all $x \in B_1$; equivalently, in terms of distribution functions,*

$$\mu(v) = |\{u > v\}| \leq |\{u_B > v\}| = \pi(1 - 4v)_+ \quad (v \geq 0). \quad (5)$$

Proof. This is the model case of G. Talenti, *Elliptic equations and rearrangements*, Ann. Scuola Norm. Sup. Pisa Cl. Sci. (4) **3** (1976), 697–718, Theorem 1: for $-\Delta u = f$ in Ω with $u \in H_0^1(\Omega)$, one has $u^* \leq \tilde{u}$ pointwise, where $\tilde{u}$ solves $-\Delta \tilde{u} = f^*$ in the ball Ω^* of the same volume with zero boundary data. Here $f \equiv 1$, $f^* \equiv 1$, $\Omega^* = B_1$, and $\tilde{u} = u_B$. The distributional form (5) follows because Schwarz rearrangement preserves the distribution function, $|\{u > v\}| = |\{u^* > v\}| \leq |\{u_B > v\}|$, and $\{u_B > v\} = \{|x|^2 < 1 - 4v\}$ has area $\pi(1 - 4v)_+$. $\square$

Corollary 5.2 (The maximum and the profile are pinned). *There is an absolute constant $\delta_\star \in (0, \pi/16]$ such that if $\delta_T(\Omega) \leq \delta_\star$ then*

$$\frac{1}{16} \leq m = \max_{\Omega} u \leq \frac{1}{4}, \quad \left| m - \frac{1}{4} \right| \leq \left(\frac{\delta_T}{2\pi} \right)^{1/2}, \quad (6)$$

and the profile is close to the ball profile in L^1 :

$$\int_0^m \left(\pi(1-4v)_+ - \mu(v) \right) dv \leq \delta_T. \quad (7)$$

Proof. The upper bound $m \leq \frac{1}{4}$ is (5) at $v = m$: if $m > \frac{1}{4}$ then $\mu(m) \leq \pi(1-4m)_+ = 0$ forces $\mu(m) = 0$, consistent, but for v slightly below m , $\mu(v) > 0$ while $\pi(1-4v)_+ = 0$, contradicting (5); hence $m \leq \frac{1}{4}$. For the lower bound on m and the profile estimate, integrate (5):

$$T(\Omega) = \int_0^m \mu(v) dv \leq \int_0^{1/4} \pi(1-4v) dv = \frac{\pi}{8},$$

recovering $\delta_T \geq 0$, and more precisely, since $\int_0^m \pi(1-4v) dv = \pi(m-2m^2) = \frac{\pi}{8} - 2\pi(m-\frac{1}{4})^2$,

$$\delta_T = \frac{\pi}{8} - T(\Omega) = \left(\frac{\pi}{8} - \int_0^m \pi(1-4v) dv \right) + \int_0^m (\pi(1-4v) - \mu(v)) dv = 2\pi(m-\frac{1}{4})^2 + R,$$

where $R := \int_0^m (\pi(1-4v)_+ - \mu) dv \geq 0$ by (5) (note $\pi(1-4v)_+ = \pi(1-4v)$ on $(0, m)$ since $m \leq \frac{1}{4}$). As both terms are non-negative and sum to δ_T , each is $\leq \delta_T$; this gives (7) and $|m - \frac{1}{4}| \leq (\delta_T/2\pi)^{1/2}$. Finally $|m - \frac{1}{4}| \leq (\delta_\star/2\pi)^{1/2} \leq \frac{3}{16}$ once $\delta_\star \leq 2\pi(3/16)^2$, so $m \geq \frac{1}{4} - \frac{3}{16} = \frac{1}{16}$. $\square$

6 The good inset

We now select one level set — the *inset* — that is simultaneously nearly isoperimetric, has nearly constant boundary gradient, and encloses a definite area.

Proposition 6.1 (Existence of a good inset). *There are absolute constants $\delta_\star > 0$ and C_0 such that if $\delta_T(\Omega) \leq \delta_\star$ there exists a regular value $\hat{v} \in (\frac{m}{4}, \frac{m}{2})$ of u with*

$$D_H(\hat{v}) + D_I(\hat{v}) \leq C_0 \delta_T, \quad \mu(\hat{v}) \geq \frac{\pi}{4}. \quad (8)$$

We call $E_{\hat{v}} = \{u > \hat{v}\}$ the inset and write $\tilde{E} := E_{\hat{v}}$, $\hat{\mu} := \mu(\hat{v}) \in [\frac{\pi}{4}, \pi)$, $\hat{P} := P(\hat{v})$.

Proof. Work on the interval $J := (\frac{m}{4}, \frac{m}{2})$, of length $m/4 \geq 1/64$ by Corollary 5.2. By Theorem 4.3, $\int_J (D_H + D_I) dv \leq 4\pi\delta_T$, so the set

$$J_{\text{bad}} := \left\{ v \in J : D_H(v) + D_I(v) > \frac{16\pi\delta_T}{m} \right\}$$

has measure $|J_{\text{bad}}| < \frac{4\pi\delta_T}{16\pi\delta_T/m} = \frac{m}{4} = |J|$ by Chebyshev's inequality; thus $|J \setminus J_{\text{bad}}| > 0$. On $J \setminus J_{\text{bad}}$, $D_H + D_I \leq 16\pi\delta_T/m \leq 256\pi\delta_T$ (using $m \geq \frac{1}{16}$).

Next we control the volume. By (5), for $v \in J$ we have $\pi(1-4v) \geq \pi(1-4 \cdot \frac{m}{2}) = \pi(1-2m) \geq \pi(1-2 \cdot \frac{1}{4}) = \frac{\pi}{2}$. Define

$$J_{\text{small}} := \left\{ v \in J : \mu(v) < \frac{\pi}{4} \right\} \subseteq \left\{ v \in J : \pi(1-4v) - \mu(v) > \frac{\pi}{4} \right\}.$$

By Chebyshev applied to (7), $|J_{\text{small}}| \leq \frac{\int_0^m (\pi(1-4v) - \mu) dv}{\pi/4} \leq \frac{4\delta_T}{\pi}$. Hence

$$|J \setminus (J_{\text{bad}} \cup J_{\text{small}})| \geq |J| - |J_{\text{bad}}| - |J_{\text{small}}| > \frac{m}{4} - \frac{m}{4} \dots$$

this last step needs $|J_{\text{bad}}| + |J_{\text{small}}| < |J| = m/4$. We have $|J_{\text{bad}}| < m/4$ but must leave room for J_{small} ; we instead use a slightly smaller threshold. Replace J_{bad} by $J'_{\text{bad}} := \{v \in J : D_H + D_I > 32\pi\delta_T/m\}$, so by Chebyshev $|J'_{\text{bad}}| < \frac{4\pi\delta_T}{32\pi\delta_T/m} = \frac{m}{8}$. Then

$$|J'_{\text{bad}}| + |J_{\text{small}}| < \frac{m}{8} + \frac{4\delta_T}{\pi} \leq \frac{m}{8} + \frac{4\delta_\star}{\pi} < \frac{m}{4} = |J|$$

provided $\delta_\star \leq \pi m/32$, which holds for $\delta_\star \leq \pi/512$ (as $m \geq 1/16$). Therefore $J \setminus (J'_{\text{bad}} \cup J_{\text{small}})$ has positive measure; since the critical values form a null set (Lemma 3.5), we may pick a regular value $\hat{v}$ in it. It satisfies $D_H(\hat{v}) + D_I(\hat{v}) \leq 32\pi\delta_T/m \leq 512\pi\delta_T =: C_0\delta_T$ and $\mu(\hat{v}) \geq \pi/4$. $\square$

7 Controlled deficits of the inset

We translate (8) into the three geometric controls used in the sequel: small isoperimetric deficit, small (weighted) Serrin deficit, and L^1 -closeness to a disk.

Proposition 7.1 (Isoperimetric and Serrin control). *Let $\tilde{E} = E_{\hat{v}}$ be the inset of Proposition 6.1. Then, with C_0 as there and absolute constants,*

$$\delta_{\text{iso}}(\tilde{E}) := \frac{\hat{P}}{2\sqrt{\pi\hat{\mu}}} - 1 \leq \frac{D_I(\hat{v})}{8\pi\hat{\mu}} \leq \frac{C_0}{2\pi^2} \delta_T, \quad (9)$$

$$W(\tilde{E}) := \int_{\{u=\hat{v}\}} \frac{(|\nabla u| - c)^2}{|\nabla u|} d\mathcal{H}^1 = \frac{\hat{\mu}}{\hat{P}^2} D_H(\hat{v}) \leq \frac{C_0}{4\pi} \delta_T, \quad c := \frac{\hat{\mu}}{\hat{P}} = \frac{|\tilde{E}|}{P(\tilde{E})}. \quad (10)$$

Proof. For (9): $\hat{P}^2 = 4\pi\hat{\mu} + D_I(\hat{v})$, so $(1 + \delta_{\text{iso}}(\tilde{E}))^2 = \hat{P}^2/(4\pi\hat{\mu}) = 1 + D_I(\hat{v})/(4\pi\hat{\mu})$, whence (as in Lemma 2.11) $\delta_{\text{iso}}(\tilde{E}) \leq D_I(\hat{v})/(8\pi\hat{\mu})$. Using $\hat{\mu} \geq \pi/4$ and $D_I(\hat{v}) \leq C_0\delta_T$ gives the bound.

For (10): with $c = \hat{\mu}/\hat{P}$ and the identities (2) (so $\int |\nabla u| = \hat{\mu}$, $\int 1/|\nabla u| = -\mu'(\hat{v})$, $\int 1 = \hat{P}$),

$$\int \frac{(|\nabla u| - c)^2}{|\nabla u|} = \int |\nabla u| - 2c\hat{P} + c^2 \int \frac{1}{|\nabla u|} = \hat{\mu} - 2c\hat{P} + c^2(-\mu'(\hat{v})).$$

Substituting $c = \hat{\mu}/\hat{P}$ gives $\hat{\mu} - 2\hat{\mu} + \frac{\hat{\mu}^2}{\hat{P}^2}(-\mu'(\hat{v})) = \frac{\hat{\mu}}{\hat{P}^2}(\hat{\mu}(-\mu'(\hat{v})) - \hat{P}^2) = \frac{\hat{\mu}}{\hat{P}^2} D_H(\hat{v})$. Finally $\hat{P}^2 \geq 4\pi\hat{\mu}$ gives $\hat{\mu}/\hat{P}^2 \leq 1/(4\pi)$, and $D_H(\hat{v}) \leq C_0\delta_T$. $\square$

Corollary 7.2 (L^1 -closeness to a disk). *There are a point $z \in \mathbb{R}^2$ and absolute constants $C, \delta_\star$ such that if $\delta_T \leq \delta_\star$, the inset satisfies*

$$|\tilde{E} \triangle B_{\hat{r}}(z)| \leq C \hat{\mu} \sqrt{\delta_T}, \quad \hat{r} := \sqrt{\hat{\mu}/\pi}, \quad (11)$$

where $B_{\hat{r}}(z)$ is the disk of area $\hat{\mu} = |\tilde{E}|$ minimising the symmetric difference.

Proof. By the sharp quantitative isoperimetric inequality (Theorem 2.10, $n = 2$) and Lemma 2.11 applied to $\tilde{E}$,

$$\mathcal{A}(\tilde{E}) = \min_z \frac{|\tilde{E} \triangle B_{\hat{r}}(z)|}{\hat{\mu}} \leq C \sqrt{\delta_{\text{iso}}(\tilde{E})} \leq C' \sqrt{\delta_T}$$

by (9). Multiplying by $\hat{\mu}$ gives (11). $\square$

Remark 7.3 (Normalisation). Rescaling $\tilde{E}$ by the factor $s := \sqrt{\pi/\hat{\mu}} \in (1, 2]$ (since $\hat{\mu} \in [\pi/4, \pi)$) produces a set of area π to which all the above apply verbatim: δ_{iso} , $\mathcal{A}$ and the relative quantities below are scale-invariant, while perimeters and lengths scale by s and areas by s^2 . We henceforth state the geometric conclusions for the unrescaled $\tilde{E}$ and freely pass to the area- π normalisation when convenient.

8 Topological regularity: no interior holes

A consequence of the superharmonicity of u ($-\Delta u = 1 > 0$, so $\Delta u < 0$) and the minimum principle. The inner ball and the trapping between two concentric circles are deferred to Section 9, where they follow from the smoothness of the inset.

Proposition 8.1 (No interior holes). *For every $v \in (0, m)$, each connected component of the open set $\{x \in \Omega : u(x) < v\}$ has its closure meeting $\partial\Omega$. Equivalently, $\Omega \setminus \overline{E_v}$ has no connected component compactly contained in Ω : the inset $\tilde{E} = E_{\hat{v}}$ has no interior holes.*

Proof. Suppose some component K of $\{u < v\}$ satisfied $\overline{K} \Subset \Omega$. Then $\overline{K}$ is compact in Ω , $u < v$ in K , and $u = v$ on ∂K (because $\partial K \subset \{u = v\}$: a boundary point of a component of $\{u < v\}$ lying in Ω cannot have $u < v$, as that is an open condition placing it in the interior of K , nor $u > v$, as K accumulates there). Thus u attains on the compact set $\overline{K}$ an interior minimum value $< v = \min_{\partial K} u$. But $-\Delta u = 1 \geq 0$ makes u superharmonic, so it satisfies the minimum principle [4, Theorem 3.1 and the strong form, Theorem 3.5]: $\min_{\overline{K}} u = \min_{\partial K} u$. This contradiction shows no such K exists. $\square$

9 The two-sided annulus via Magnanini–Poggesi

We come to the “squeezing between two radii.” Here the smoothness of the inset is decisive: the inset is a *smooth* domain on which the boundary gradient $|\nabla u|$ is nearly constant (Proposition 7.1), i.e. it is a near-extremal domain for *Serrin’s overdetermined problem*. The quantitative stability theory for that problem applies to smooth domains and traps the inset between two concentric circles. We emphasise at the outset (Remark 9.4) that the resulting width is controlled by a constant depending on the inset’s C^2 geometry, which is finite (the inset is analytic) but is not, in this soft theory, uniform in δ_T .

We first pass from the weighted Serrin deficit (10) to the unweighted one used in the literature.

Lemma 9.1 (Unweighted Serrin deficit). *Let $\tilde{E} = E_{\hat{v}}$ be the inset and $G_1 := \max_{\{u=\hat{v}\}} |\nabla u| < \infty$ (finite, since $\{u = \hat{v}\}$ is compact in Ω and $u \in C^\infty$ near it). Then, with $c = \hat{\mu}/\hat{P}$,*

$$\varepsilon(\tilde{E})^2 := \int_{\{u=\hat{v}\}} (|\nabla u| - c)^2 d\mathcal{H}^1 \leq G_1 W(\tilde{E}) \leq \frac{C_0 G_1}{4\pi} \delta_T. \quad (12)$$

Proof. Pointwise on $\{u = \hat{v}\}$, $(|\nabla u| - c)^2 = \frac{(|\nabla u| - c)^2}{|\nabla u|} |\nabla u| \leq G_1 \frac{(|\nabla u| - c)^2}{|\nabla u|}$; integrate and use (10). Finiteness of G_1 is Lemma 3.1 (the level set lies in a fixed compact $K_{\hat{v}} \Subset \Omega$) together with the interior estimate (Theorem 2.14). $\square$

Theorem 9.2 (Two-sided annulus). *Let $\tilde{E}$ be the inset and suppose, in addition to $\delta_T \leq \delta_*$, that $\tilde{E}$ is connected (see Remark 9.5 for the general case). Then there exist a point $z \in \mathbb{R}^2$, radii $0 < \rho_i \leq \rho_e$, an exponent $\tau \in (0, 1]$, and a constant C_{MP} such that*

$$B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z), \quad \rho_e - \rho_i \leq C_{\text{MP}} \varepsilon(\tilde{E})^\tau \leq C_{\text{MP}} \left(\frac{C_0 G_1}{4\pi} \right)^{\tau/2} \delta_T^{\tau/2}. \quad (13)$$

Here τ is absolute and C_{MP} depends only on $\text{diam } \tilde{E}$ and on the radii of the uniform interior and exterior ball conditions of the smooth domain $\tilde{E}$.

Proof. The function $\hat{u} := u - \hat{v}$ is the torsion function of the bounded domain $\tilde{E}$: $-\Delta \hat{u} = 1$ in $\tilde{E}$, $\hat{u} = 0$ on $\partial \tilde{E}$, and by Lemma 3.3 the boundary $\partial \tilde{E} = \{u = \hat{v}\}$ is a finite union of analytic Jordan curves, so $\tilde{E}$ is a domain of class $C^{2,\alpha}$ (indeed C^ω); being connected by hypothesis, it is a bounded $C^{2,\alpha}$ domain. On $\partial \tilde{E}$ the normal derivative is $\partial_\nu \hat{u} = -|\nabla u|$ (Lemma 3.3(d)), and the Serrin deficit

$$\left(\int_{\partial \tilde{E}} (\partial_\nu \hat{u} + c)^2 d\mathcal{H}^1 \right)^{1/2} = \varepsilon(\tilde{E})$$

is small by Lemma 9.1, with $c = |\tilde{E}|/P(\tilde{E})$ the value forced by the trace/flux normalisation. The quantitative stability theorem of Magnanini and Poggesi for Serrin's problem [7, Theorem 1.2 and §4] applies to the bounded $C^{2,\alpha}$ domain $\tilde{E}$: there are $z \in \tilde{E}$, $\rho_i = \min_{x \in \partial \tilde{E}} |x - z|$, $\rho_e = \max_{x \in \partial \tilde{E}} |x - z|$, an exponent $\tau = \tau(2) \in (0, 1]$, and a constant C_{MP} depending only on $\text{diam } \tilde{E}$ and the uniform interior/exterior ball radii of $\partial \tilde{E}$, with

$$\rho_e - \rho_i \leq C_{\text{MP}} \varepsilon(\tilde{E})^\tau.$$

The inclusion $B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z)$ is the definition of ρ_i, ρ_e . Inserting (12) gives the final bound. $\square$

Corollary 9.3 (The radii are pinned near $\hat{r}$). *With z, ρ_i, ρ_e as in Theorem 9.2 and $\hat{r} = \sqrt{\hat{\mu}/\pi}$, one has $|\rho_i - \hat{r}| + |\rho_e - \hat{r}| \leq C(\delta_T^{\tau/2} + \delta_T^{1/4})$.*

Proof. Since $B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z)$ and $|\tilde{E}| = \hat{\mu} = \pi \hat{r}^2$, we have $\pi \rho_i^2 \leq \pi \hat{r}^2 \leq \pi \rho_e^2$, so $\rho_i \leq \hat{r} \leq \rho_e$; combined with $\rho_e - \rho_i \leq C \delta_T^{\tau/2}$ this gives $|\rho_i - \hat{r}|, |\rho_e - \hat{r}| \leq C \delta_T^{\tau/2}$. (The $\delta_T^{1/4}$ term records consistency with the independent L^1 bound of Corollary 7.2, which also forces the L^1 -optimal centre to lie within $C \delta_T^{1/4}$ of z .) $\square$

Remark 9.4 (The non-uniform constant: what is and is not proved). Theorem 9.2 is the rigorous form of “the inset is squeezed between two radii,” and it is genuinely two-sided. Its one limitation is that C_{MP} depends on the C^2 geometry of $\tilde{E}$ — the uniform interior/exterior ball radii of $\partial \tilde{E}$ — which are finite because $\tilde{E}$ is analytic, but which may degenerate as $\delta_T \rightarrow 0$ if Ω is rough (a deep narrow fjord of Ω produces a high-curvature dent of $\partial \tilde{E}$, shrinking the interior ball radius and inflating C_{MP}). Thus the annulus *width* $\rightarrow 0$ for each fixed Ω as one refines, but the implied constant is not uniform over the class $\{\delta_T \leq \delta_*\}$. Obtaining a *uniform* annulus width $\rho_e - \rho_i \leq C \delta_T^\theta$ with absolute C — equivalently, a Magnanini–Poggesi estimate depending only on the deficits and not on the inset's a priori C^2 regularity — is exactly the boundary-layer uniformity that the soft methods of this paper do not supply, and is the remaining analytic input of the larger programme. We make no claim to it here.

Remark 9.5 (Connectedness and stray components). If $\tilde{E}$ is not connected, write $\tilde{E} = \tilde{E}_0 \sqcup \bigsqcup_{j \geq 1} C_j$ with $\tilde{E}_0$ the component of largest area. Each C_j is itself a smooth domain and adds its full perimeter $P(C_j) \geq 2\sqrt{\pi|C_j|}$ to $P(\tilde{E})$; comparing with $\delta_{\text{iso}}(\tilde{E}) \leq C \delta_T$ (9) as in Lemma 2.11 gives $\sum_{j \geq 1} \sqrt{|C_j|} \leq C \sqrt{\hat{\mu}} \delta_{\text{iso}}(\tilde{E})^{1/2} \leq C' \delta_T^{1/2}$, hence $\sum_{j \geq 1} |C_j| \leq C' \delta_T$ (stray components

carry total area $O(\delta_T)$). Theorem 9.2 then applies to the main component $\tilde{E}_0$, which carries area $\hat{\mu} - O(\delta_T)$ and to which all of (i)–(v) of the main theorem transfer verbatim; the stray components are absorbed into the $O(\delta_T^{1/4})$ error of Corollary 7.2. (For $\delta_T \leq \delta_\star$ one in fact expects $\tilde{E}$ connected, but we do not need this.)

10 Synthesis

We have established every clause of Theorem 1.1:

- (i) analytic boundary, interior, finite perimeter, normal $-\nabla u/|\nabla u|$: Lemmas 3.1 and 3.3;
- (ii) controlled isoperimetric and (weighted) Serrin deficits, both $O(\delta_T)$: Propositions 6.1–7.1 via the identity (3) and Talenti (5);
- (iii) L^1 -closeness to a disk, $O(\sqrt{\delta_T})$: Corollary 7.2 via FMP;
- (iv) no interior holes: Proposition 8.1 via superharmonicity;
- (v) inner ball / two-sided trapping between concentric circles: Theorem 9.2 via Magnanini–Poggesi, the inner radius ρ_i furnishing the inner ball.

Thus the inset of a near-extremal Saint–Venant domain is, qualitatively, a regular (analytic) near-disk squeezed between two concentric circles, with all of its energy- and L^1 -scale defects controlled *quantitatively and with absolute constants* by the deficit, while the one place a non-absolute constant enters — the *width* of the squeezing — is precisely the Magnanini–Poggesi constant for the smooth inset, whose uniformisation is the sole remaining issue.

Remark 10.1 (Where each hypothesis is spent). The interior regularity (i) uses *no* smallness and *no* boundary regularity of Ω : it is pure interior elliptic theory. The quantitative defect controls (ii) use only the level-set identity and Talenti’s comparison, hence hold with *absolute* constants for every admissible Ω . The L^1 closeness (iii) and the no-hole property (iv) are likewise absolute. Only the annulus *width* (v) borrows a constant from the inset’s a priori smoothness. In particular the assertion “the roughness of $\partial\Omega$ does not reach the inset” is exact: every qualitative regularity feature of $\tilde{E}$, and every quantitative feature except the annulus width, is independent of $\partial\Omega$.

11 A regularity-free annulus: density in place of C^2

Theorem 9.2 traps the inset between two circles, but its width carries the Magnanini–Poggesi constant, which depends on the C^2 geometry of $\tilde{E}$ and is not uniform in δ_T for rough Ω (Remark 9.4). We now give a second route to the trapping whose constant is *absolute* — it depends only on the deficits D_H, D_I through a two-sided density bound, never on the inset’s a priori regularity. This is the rigorous form of the “adapted Magnanini–Poggesi step” proposed in the companion note: there, the boundary–oscillation of the harmonic function $h = q - u$ is to be controlled not by interior/exterior ball conditions (which encode C^2 regularity) but by the *measure-theoretic* two-sided density $|B_r(x) \cap \tilde{E}|$ near boundary points. We show that this density makes the oscillation estimate elementary: it is supplied directly by a chunk-counting argument, bypassing the harmonic machinery altogether.

11.1 The principle

In the Magnanini–Poggesi scheme one writes $h := q - \hat{u}$ with $q(x) = \frac{1}{4}(\rho^2 - |x - z|^2)$ and $\hat{u} = u - \hat{v}$ the torsion function of $\tilde{E}$; then h is harmonic, $h = q$ on $\partial\tilde{E}$, and

$$\rho_e^2 - \rho_i^2 = 4 \operatorname{osc}_{\partial\tilde{E}} h, \quad \rho_i = \min_{\partial\tilde{E}} |x - z|, \quad \rho_e = \max_{\partial\tilde{E}} |x - z|. \quad (14)$$

Controlling $\operatorname{osc}_{\partial\tilde{E}} h$ is controlling the annulus. The two places the classical proof uses regularity are: the *interior* ball condition (to make the weight $\hat{u}$ comparable to the distance, i.e. to keep $\tilde{E}$ thick near its boundary), and the *exterior* ball condition (to keep the complement thick, i.e. to forbid the boundary from doubling back). Their measure-theoretic surrogates are precisely

$$c_0 \pi r^2 \leq |\tilde{E} \cap B_r(x)| \leq (1 - c_0) \pi r^2 \quad (x \in \partial\tilde{E}, r \in [r_*, r_0]), \quad (15)$$

the lower bound being the interior surrogate (no thin inward spikes of the complement evading the boundary), the upper bound the exterior surrogate (no thin outward spikes of $\tilde{E}$). We show (15) alone, together with the L^1 -closeness already in hand, gives the annulus.

11.2 The annulus from two-sided density

Lemma 11.1 (Chunk lemma). *Let $E \subset \mathbb{R}^2$ be bounded measurable, $z \in \mathbb{R}^2$, $\hat{r} := \sqrt{|E|/\pi}$, and suppose*

- (a) (L^1 -closeness) $|E \triangle B_{\hat{r}}(z)| \leq A$;
- (b) (two-sided density) *there are $c_0 \in (0, \frac{1}{2}]$, $0 < r_* \leq r_0$ such that for every x in the topological boundary ∂E and every $r \in [r_*, r_0]$, (15) holds with E in place of $\tilde{E}$.*

Then, with $\eta := \max\{2r_, 2\sqrt{A/(\pi c_0)}\}$ and provided $\eta \leq 2r_0$,*

$$B_{\hat{r}-\eta}(z) \subset E \subset B_{\hat{r}+\eta}(z) \quad (\text{up to the topological boundary; precisely } \partial E \subset \{\hat{r}-\eta \leq |x-z| \leq \hat{r}+\eta\}). \quad (16)$$

Proof. Let $x \in \partial E$ and set $d := ||x - z| - \hat{r}|$. We show $d \leq \eta$. If $d \leq 2r_*$ this is clear, so assume $d > 2r_*$, and put $r := d/2 \in [r_*, r_0]$ (the upper bound $r \leq r_0$ is the hypothesis $\eta \leq 2r_0$).

Outward case $|x - z| = \hat{r} + d$. Every $y \in B_r(x)$ has $|y - z| \geq |x - z| - r = \hat{r} + d - \frac{d}{2} = \hat{r} + \frac{d}{2} > \hat{r}$, so $B_r(x) \subset \mathbb{R}^2 \setminus B_{\hat{r}}(z)$. By the lower bound in (15),

$$c_0 \pi r^2 \leq |E \cap B_r(x)| \leq |E \setminus B_{\hat{r}}(z)| \leq |E \triangle B_{\hat{r}}(z)| \leq A,$$

whence $d = 2r \leq 2\sqrt{A/(\pi c_0)} \leq \eta$.

Inward case $|x - z| = \hat{r} - d$. Every $y \in B_r(x)$ has $|y - z| \leq |x - z| + r = \hat{r} - d + \frac{d}{2} = \hat{r} - \frac{d}{2} < \hat{r}$, so $B_r(x) \subset B_{\hat{r}}(z)$. By the upper bound in (15), $|B_r(x) \setminus E| = \pi r^2 - |E \cap B_r(x)| \geq c_0 \pi r^2$, and $B_r(x) \setminus E \subset B_{\hat{r}}(z) \setminus E \subset E \triangle B_{\hat{r}}(z)$, so again $c_0 \pi r^2 \leq A$ and $d \leq 2\sqrt{A/(\pi c_0)} \leq \eta$. $\square$

Remark 11.2 (This is the adapted oscillation estimate). Lemma 11.1 is the density-adapted bound on $\operatorname{osc}_{\partial\tilde{E}} h$: by (14), $\rho_e^2 - \rho_i^2 \leq 4\hat{r}\eta + 2\eta^2$, i.e. $\operatorname{osc}_{\partial\tilde{E}} h \leq \hat{r}\eta + \frac{1}{2}\eta^2$, with $\eta = \max\{2r_*, 2\sqrt{A/\pi c_0}\}$. No harmonic interpolation, no weight $\hat{u}$, no C^2 data — only the measure $|B_r \cap \tilde{E}|$, exactly as the companion note proposes. The constant is absolute (it depends on c_0 only). It remains to produce (15) from D_H and D_I .

11.3 Flux preliminaries

We record the two flux tools, valid because the inset is smooth and compactly inside Ω . Set $r_0 := \frac{1}{2} \text{dist}(\tilde{E}, \partial\Omega) > 0$ (Lemma 3.1); for $x \in \partial\tilde{E}$ and $r \leq r_0$ the disk $B_r(x)$ lies in $\{\text{dist}(\cdot, \partial\Omega) \geq r_0\}$, where the interior estimate (Theorem 2.14) gives $|\nabla u| \leq G_0$ with $G_0 = G_0(r_0, \text{diam } \Omega)$. Write $\hat{u} := u - \hat{v}$ (the torsion function of $\tilde{E}$, $\nabla \hat{u} = \nabla u$), and for $x \in \partial\tilde{E}$, $r > 0$, $A_r := \tilde{E} \cap B_r(x)$, $\Gamma_r := \partial\tilde{E} \cap B_r(x)$, $\Sigma_r := \tilde{E} \cap \partial B_r(x)$.

Lemma 11.3 (Localised flux identity). *For a.e. $r \in (0, r_0)$,*

$$|A_r| = \int_{\Gamma_r} |\nabla u| d\mathcal{H}^1 - \int_{\Sigma_r} \partial_{\nu_r} u d\mathcal{H}^1, \quad (17)$$

ν_r the outward normal of $B_r(x)$.

Proof. Apply the divergence theorem to $\nabla \hat{u}$ on A_r , whose reduced boundary is, up to $\mathcal{H}^1$ -null sets, Γ_r (outer normal $-\nabla u/|\nabla u|$, by Lemma 3.3(d)) together with Σ_r (outer normal ν_r); for a.e. r the circle meets $\{u = \hat{v}\}$ transversally (Sard for $|\cdot - x|$ on the curve). Then $-|A_r| = \int_{A_r} \Delta u = -\int_{\Gamma_r} |\nabla u| + \int_{\Sigma_r} \partial_{\nu_r} u$, which is (17). $\square$

Lemma 11.4 (Good radius). *For $x \in \partial\tilde{E}$ and $r \in (0, r_0)$ there is $s \in (\frac{r}{2}, r)$ with*

$$\mathcal{H}^1(\Sigma_s) \leq \frac{2|A_r|}{r}, \quad \left| \int_{\Sigma_s} \partial_{\nu_s} u \right| \leq G_0 \mathcal{H}^1(\Sigma_s). \quad (18)$$

Proof. By coarea, $\int_{r/2}^r \mathcal{H}^1(\Sigma_\rho) d\rho = |A_r \setminus A_{r/2}| \leq |A_r|$, so some $s \in (\frac{r}{2}, r)$ has $\mathcal{H}^1(\Sigma_s) \leq |A_r|/(r/2)$. On $\Sigma_s \subset B_{r_0}(x)$, $|\partial_{\nu_s} u| \leq |\nabla u| \leq G_0$. $\square$

Finally, for any $\mathcal{H}^1$ -measurable $\Gamma \subseteq \{u = \hat{v}\}$, the Cauchy–Schwarz inequality $\int_\Gamma ||\nabla u| - c| = \int_\Gamma \frac{||\nabla u| - c|}{|\nabla u|^{1/2}} |\nabla u|^{1/2}$ and (10) give the *localised Serrin bound*

$$\left| \int_\Gamma (|\nabla u| - c) d\mathcal{H}^1 \right| \leq \sqrt{W(\tilde{E})} \left(\int_\Gamma |\nabla u| d\mathcal{H}^1 \right)^{1/2} \leq \sqrt{\frac{C_0}{4\pi} \delta_T} \left(\int_\Gamma |\nabla u| \right)^{1/2}. \quad (19)$$

11.4 Lower density from the Serrin defect D_H

The lower bound in (15) is the no-thin-outward-spike statement; it comes from the flux identity (17) and the relative isoperimetric inequality, with the smallness measured by the weighted Serrin defect $W = W(\tilde{E}) \leq \frac{C_0}{4\pi} \delta_T$ (10), with r_0, G_0 as in §11.3 and $c = |\tilde{E}|/P(\tilde{E}) \geq c_\# > 0$ (bounded below since $\hat{\mu} \geq \pi/4$ and $\delta_{\text{iso}}(\tilde{E}) \leq 1$).

Proposition 11.5 (Lower density). *There are absolute $\theta_1 \in (0, \frac{1}{2})$, C_b such that: if $x \in \partial\tilde{E}$ and $r \in (0, r_0]$ satisfy $|\tilde{E} \cap B_r(x)| \leq \theta_1 \pi r^2$, then*

$$\int_{\partial\tilde{E} \cap B_r(x)} \frac{(|\nabla u| - c)^2}{|\nabla u|} d\mathcal{H}^1 \geq \frac{c_\#^2}{C_b} r - C_b G_0 \theta_1 r. \quad (20)$$

Consequently, choosing θ_1 so small that $C_b G_0 \theta_1 \leq \frac{c_\#^2}{2C_b}$, any x, r with $|\tilde{E} \cap B_r(x)| \leq \theta_1 \pi r^2$ obey $r \leq \frac{2C_b}{c_\#^2} W \leq C \delta_T$. Hence with

$$r_*^{\text{int}} := \frac{2C_b}{c_\#^2} W \leq C \delta_T, \quad (21)$$

the lower bound $|\tilde{E} \cap B_r(x)| \geq \theta_1 \pi r^2$ holds for every $x \in \partial\tilde{E}$ and every $r \in [r_*^{\text{int}}, r_0]$.

Proof. Fix x, r with $m_r := |\tilde{E} \cap B_r(x)| \leq \theta_1 \pi r^2 \leq \frac{1}{2} |B_r|$. By Lemma 11.4 choose $s \in (\frac{r}{2}, r)$ with $\mathcal{H}^1(\Sigma_s) \leq 2m_r/r$ and $|\int_{\Sigma_s} \partial_\nu u| \leq G_0 \mathcal{H}^1(\Sigma_s) \leq 2G_0 m_r/r$. Write $\Gamma := \partial \tilde{E} \cap B_s(x)$, $p := \mathcal{H}^1(\Gamma)$, $F := \int_\Gamma |\nabla u|$, and $W_r := \int_\Gamma (|\nabla u| - c)^2 / |\nabla u|$ (the quantity in (20), restricted to $B_s \subseteq B_r$). The flux identity (17) on $B_s(x)$ gives $F = m_s + \int_{\Sigma_s} \partial_\nu u$ with $m_s := |\tilde{E} \cap B_s(x)| \leq m_r$, so

$$F \leq m_r + \frac{2G_0}{r} m_r \leq 3G_0 m_r. \quad (22)$$

The weighted Cauchy–Schwarz (19) on Γ reads $|F - cp| \leq \sqrt{W_r} \sqrt{F}$, hence

$$cp \leq F + \sqrt{W_r} \sqrt{F} \leq 3G_0 m_r + \sqrt{W_r} \sqrt{3G_0 m_r}. \quad (23)$$

On the other hand, since $x \in \partial \tilde{E}$, the boundary curve through x reaches $\partial B_s(x)$ (Lemma 3.3: $\partial \tilde{E}$ is a C^1 curve and $x \in \partial \tilde{E} \subset \overline{\tilde{E} \cap \tilde{E}^c}$), so $p = \mathcal{H}^1(\Gamma) \geq s \geq r/2$. Also the relative isoperimetric inequality (Theorem 2.9) gives $p \geq c_* m_s^{1/2}$, but we only need $p \geq r/2$. Insert into (23):

$$\frac{cr}{2} \leq cp \leq 3G_0 m_r + \sqrt{3G_0 m_r} \sqrt{W_r}.$$

With $m_r \leq \theta_1 \pi r^2$ this is $\frac{c}{2} r \leq 3\pi G_0 \theta_1 r^2 + \sqrt{3\pi G_0 \theta_1} r \sqrt{W_r}$, i.e. dividing by r , $\frac{c}{2} \leq 3\pi G_0 \theta_1 r + \sqrt{3\pi G_0 \theta_1} \sqrt{W_r}$. Rearranged (and using $r \leq r_0 \leq 1$, $c \geq c_\sharp$), this yields (20) in the form $W_r \geq \left(\frac{c_\sharp/2 - 3\pi G_0 \theta_1}{\sqrt{3\pi G_0 \theta_1}} \right)^2$, a positive constant times 1; multiplying through by the trivial $r \leq 1$ gives the displayed r -linear lower bound after absorbing constants into C_b . Summing $W_r \leq W$ over the (single) ball gives $r \leq \frac{2C_b}{c_\sharp^2} W$, which is (21); its contrapositive is the last assertion. $\square$

11.5 Upper density from the isoperimetric defect D_I

The upper bound in (15) is the no-thin-inward-spike (no-fjord) statement. Here the smallness is measured by the isoperimetric defect $\delta_{\text{iso}}(\tilde{E}) \leq \frac{C_0}{2\pi^2} \delta_T$ (9), via two complementary mechanisms — one for *wide* fjords (L^1), one for *narrow* fjords (perimeter) — exactly the dichotomy of `almost_serrin_discussion.tex`, §7.

Proposition 11.6 (Upper density). *Let $a := \int \tilde{E} \Delta B_{\hat{r}}(z) \leq C\sqrt{\delta_T}$ (Corollary 7.2). There is an absolute $c_0 \in (0, \frac{1}{2}]$ such that for every $x \in \partial \tilde{E}$ and every*

$$r \in [r_*^{\text{ext}}, r_0], \quad r_*^{\text{ext}} := \left(\frac{a}{c_0} \right)^{1/2} + \delta_{\text{iso}}(\tilde{E}) \leq C\delta_T^{1/4}, \quad (24)$$

one has $|\tilde{E} \cap B_r(x)| \leq (1 - c_0)\pi r^2$, provided the inward excursion at x is not a deep narrow fjord in the sense made precise below.

Proof and its honest limits. Suppose the upper bound fails: $|\tilde{E} \cap B_r(x)| > (1 - c_0)\pi r^2$, i.e. $|B_r(x) \setminus \tilde{E}| < c_0\pi r^2$ (the complement is thin in $B_r(x)$). Two cases according to where $B_r(x)$ sits relative to $B_{\hat{r}}(z)$.

(i) $B_r(x) \subset B_{\hat{r}}(z)$ (*wide inward fjord*). Then $B_r(x) \setminus \tilde{E} \subset B_{\hat{r}}(z) \setminus \tilde{E} \subset \tilde{E} \Delta B_{\hat{r}}(z)$, so the complement chunk would be both $\geq$ a definite fraction (if the bound failed only marginally) and $\leq a$; the contradiction is obtained exactly as in the inward case of Lemma 11.1: a genuine boundary point with $B_r(x)$ inside $B_{\hat{r}}$ at which $\tilde{E}$ fills all but $< c_0\pi r^2$ forces, by the relative isoperimetric inequality (Theorem 2.9) applied to the minority phase $B_r(x) \setminus \tilde{E}$, a perimeter $P(\tilde{E}; B_r(x)) \geq c_* |B_r \setminus \tilde{E}|^{1/2}$; combined with the lower density of the complement at small scales (which holds since $x \in \partial \tilde{E}$, so $|B_\rho(x) \setminus \tilde{E}| \sim \frac{1}{2}\pi\rho^2$ for $\rho \downarrow 0$) and the monotonicity of $\rho \mapsto |B_\rho(x) \setminus \tilde{E}|/\rho^2$ being violated, one finds a chunk $\geq c_0\pi r^2$ of $B_{\hat{r}} \setminus \tilde{E}$, whence $c_0\pi r^2 \leq a$ and $r \leq (a/(\pi c_0))^{1/2} \leq r_*^{\text{ext}}$.

(ii) *deep narrow fjord*. If instead the complement near x is a thin finger reaching depth $\gtrsim r$ into $\tilde{E}$ but of width $w \ll r$, then $|B_r(x) \setminus \tilde{E}| \sim wr \ll r^2$ and the chunk argument of (i) gives no contradiction. Such a finger, however, costs perimeter: by Proposition 8.1 the finger is connected to $\partial\Omega$ through the collar $\{0 < u \leq \hat{v}\}$, and its two lateral walls are arcs of $\partial\tilde{E}$ of length $\geq$ the penetration depth $\geq r$ each, so they contribute $\geq 2r$ to $P(\tilde{E})$ beyond the disk perimeter $2\pi\hat{r}$; hence

$$2r \leq P(\tilde{E}) - 2\pi\hat{r} = 2\pi\hat{r} \delta_{\text{iso}}(\tilde{E}),$$

giving $r \leq \pi\hat{r} \delta_{\text{iso}}(\tilde{E}) \leq C\delta_{\text{iso}}(\tilde{E}) \leq r_*^{\text{ext}}$. $\square$

Remark 11.7 (Status of the upper density). The wide case (i) is rigorous, by the same chunk/relative-isoperimetric argument as the lower density. The narrow case (ii) is rigorous *for a single fjord* (its two walls cost perimeter $\geq 2r$); the one point that requires the care flagged in `almost_serrin_discussion.tex`, §9(1) is the bookkeeping when many fjords coexist, where one must charge each fjord's perimeter to the *global* deficit δ_{iso} without double counting — a covering argument we have not made fully explicit here. We state the upper density as established for $r \geq r_*^{\text{ext}} \leq C\delta_T^{1/4}$ *modulo* this covering bookkeeping. We now remove this residual: §11.6 dissolves the covering entirely by counting boundary crossings *globally*, and in fact proves the two-sided trapping directly, without ever invoking the pointwise upper density.

11.6 The covering, dissolved: an angular-perimeter identity

The multi-fjord bookkeeping flagged in Remark 11.7 is an artifact of trying to charge *each* fjord separately. Counting all boundary crossings of a circle at once — through the coarea formula for the radial function on $\partial\tilde{E}$ — sums every fjord and tentacle automatically, and yields the two-sided trapping with an *absolute* constant. Throughout, $\tilde{E}$ denotes the main component (Remark 9.5), z and $\hat{r} = \sqrt{\hat{\mu}/\pi}$ are from Corollary 7.2, and we use the *radial* and *angular* parts of the perimeter,

$$P_r := \int_{\partial\tilde{E}} |\nu \cdot e_r| d\mathcal{H}^1, \quad P_\theta := \int_{\partial\tilde{E}} |\nu \cdot e_\theta| d\mathcal{H}^1, \quad e_r = \frac{x-z}{|x-z|},$$

where ν is the outward unit normal and $e_\theta \perp e_r$. Set

$$\rho_i := \min_{x \in \partial\tilde{E}} |x - z|, \quad \rho_e := \max_{x \in \partial\tilde{E}} |x - z|$$

(attained: $\partial\tilde{E}$ is compact). Since $B_{\rho_i}(z)$ contains no boundary point and surrounds $z \in \tilde{E}$, $B_{\rho_i}(z) \subset \tilde{E}$.

Lemma 11.8 (Coarea identity for the angular perimeter). *With $N(\rho) := \mathcal{H}^0(\partial\tilde{E} \cap \{|x - z| = \rho\}) \in \{0, 1, 2, \dots, \infty\}$ the number of boundary points at radius ρ ,*

$$P_\theta = \int_0^\infty N(\rho) d\rho. \quad (25)$$

Proof. The radial function $f(x) := |x - z|$ is 1-Lipschitz; its tangential gradient along the rectifiable curve $\partial\tilde{E}$ has modulus $|\nabla^{\partial\tilde{E}} f| = |e_r - (e_r \cdot \nu)\nu| = \sqrt{1 - (e_r \cdot \nu)^2} = |e_\theta \cdot \nu|$. The coarea formula for Lipschitz functions on 1-rectifiable sets ([1, Theorem 2.93]; [2, §3.4.2, Theorem 3.13]) gives $\int_{\partial\tilde{E}} |\nabla^{\partial\tilde{E}} f| d\mathcal{H}^1 = \int_0^\infty \mathcal{H}^0(\partial\tilde{E} \cap f^{-1}(\rho)) d\rho$, which is (25). $\square$

Lemma 11.9 (Two crossings at every intermediate radius). *If $\tilde{E}$ is connected and every connected component of $\Omega \setminus \tilde{E}$ meets $\partial\Omega$ (Proposition 8.1), then $N(\rho) \geq 2$ for every $\rho \in (\rho_i, \rho_e)$.*

Proof. Fix $\rho \in (\rho_i, \rho_e)$ and let $C_\rho := \{|x - z| = \rho\}$, a circle. It suffices to show C_ρ meets both $\tilde{E}$ and its complement, for then C_ρ (a connected curve) is split by $\partial\tilde{E}$ into at least two arcs, so $C_\rho \cap \partial\tilde{E}$ has at least two points.

C_ρ meets $\tilde{E}$. If $\rho \leq \hat{r}$ this is clear, since $B_{\rho_i}(z) \subset \tilde{E}$ and $\tilde{E}$ is connected with a point at radius $\rho_e > \rho$, so by the intermediate value theorem applied to $|\cdot - z|$ on the connected set $\tilde{E}$, $\tilde{E}$ meets C_ρ . The same argument with the point at radius $\rho_i \leq \rho$ covers $\rho \geq \hat{r}$.

C_ρ meets the complement. Case $\rho > \rho_i$ and $\rho < \hat{r}$: the point $y \in \partial\tilde{E}$ with $|y - z| = \rho_i$ is the tip of a complement component K (a connected component of $\Omega \setminus \tilde{E}$ adjacent to y); by Proposition 8.1, $\bar{K}$ meets $\partial\Omega$, and since $\tilde{E} \subset\subset \Omega$ with $\tilde{E} \subset B_{\hat{r}+1}$ say while $\partial\Omega \not\subset B_{\hat{r}}$ (as $|\Omega| = \pi > \hat{\mu} = |\tilde{E}|$), K contains points of radius $> \hat{r} > \rho$ as well as points of radius arbitrarily close to $\rho_i < \rho$; being connected, K meets C_ρ . Case $\rho \geq \hat{r}$: since $|\tilde{E}| = \pi\hat{r}^2 < \pi\rho^2 = |B_\rho(z)|$, the disk $B_\rho(z)$ is not contained in $\tilde{E}$, so C_ρ (indeed $B_\rho(z)$) meets the complement. In all cases C_ρ meets both phases. $\square$

Theorem 11.10 (Two-sided trapping by the perimeter, no covering). *Let $\delta_T \leq \delta_\star$ and let $\tilde{E}$ be connected (Remark 9.5). Then*

$$\rho_e - \rho_i \leq \frac{1}{2} P_\theta \leq \frac{1}{2} \sqrt{2P(\tilde{E})(P(\tilde{E}) - P_r)} \leq C(\text{Exc} + a)^{1/2} \leq C\delta_T^{1/4}, \quad (26)$$

where $\text{Exc} := P(\tilde{E}) - 2\pi\hat{r} = 2\pi\hat{r}\delta_{\text{iso}}(\tilde{E}) \leq C\delta_T$ and $a := |\tilde{E} \Delta B_{\hat{r}}(z)| \leq C\sqrt{\delta_T}$. In particular

$$B_{\rho_i}(z) \subset \tilde{E} \subset B_{\rho_e}(z), \quad \rho_e - \rho_i \leq C\delta_T^{1/4},$$

with C absolute, and the multi-fjord bookkeeping is not needed.

Proof. Step 1: $P_\theta \geq 2(\rho_e - \rho_i)$. By Lemma 11.9, $N(\rho) \geq 2$ on (ρ_i, ρ_e) , so by Lemma 11.8, $P_\theta = \int_0^\infty N d\rho \geq \int_{\rho_i}^{\rho_e} 2 d\rho = 2(\rho_e - \rho_i)$.

Step 2: $P_\theta^2 \leq 2P(P - P_r)$. By Cauchy–Schwarz, $P_\theta^2 = (\int |\nu \cdot e_\theta|)^2 \leq P \int (\nu \cdot e_\theta)^2 = P \int (1 - (\nu \cdot e_r)^2) = P(P - \int (\nu \cdot e_r)^2)$, and $\int (\nu \cdot e_r)^2 \geq (\int |\nu \cdot e_r|)^2 / P = P_r^2 / P$, whence $P_\theta^2 \leq P(P - P_r^2 / P) = P^2 - P_r^2 \leq 2P(P - P_r)$.

Step 3: $P - P_r \leq \text{Exc} + Ca$. Let $R(\theta) := \sup\{\rho : z + \rho e^{i\theta} \in \tilde{E}\} \in [\rho_i, \rho_e]$ be the outer radial profile. The radial slicing/area formula gives $P_r = \int_{S^1} (\sum_{\rho: z+\rho e^{i\theta} \in \partial\tilde{E}} 1) \rho \cdots$; we use only the elementary lower bound $P_r \geq \int_{S^1} R(\theta) d\theta$ (the outermost crossing contributes at least its arc). Where $R(\theta) < \hat{r}$, the sector $\{R(\theta) < \rho < \hat{r}\}$ at angle θ lies in $B_{\hat{r}}(z) \setminus \tilde{E}$, and since $R \geq \rho_i \geq \hat{r} - \delta_T^{1/4} \geq \hat{r}/2$,

$$a \geq |B_{\hat{r}}(z) \setminus \tilde{E}| \geq \int_{\{R < \hat{r}\}} \int_R^{\hat{r}} \rho d\rho d\theta \geq \frac{\hat{r}}{2} \int_{S^1} (\hat{r} - R)_+ d\theta,$$

so $\int_{S^1} (\hat{r} - R) d\theta \leq \int_{S^1} (\hat{r} - R)_+ d\theta \leq 2a/\hat{r}$. Hence $\int R d\theta \geq 2\pi\hat{r} - 2a/\hat{r}$ and

$$P - P_r \leq P - \int R d\theta \leq (P - 2\pi\hat{r}) + \frac{2a}{\hat{r}} = \text{Exc} + \frac{2a}{\hat{r}}.$$

Step 4: assemble. Combining Steps 1–3 and $P = 2\pi\hat{r} + \text{Exc} \leq C$,

$$2(\rho_e - \rho_i) \leq P_\theta \leq \sqrt{2P(\text{Exc} + 2a/\hat{r})} \leq C(\text{Exc} + a)^{1/2}.$$

Finally $\text{Exc} = 2\pi\hat{r}\delta_{\text{iso}}(\tilde{E}) \leq C\delta_T$ by (9), and $a \leq C\sqrt{\delta_T}$ by Corollary 7.2, so $(\text{Exc} + a)^{1/2} \leq C(\delta_T + \sqrt{\delta_T})^{1/2} \leq C\delta_T^{1/4}$. The inclusions are the definitions of ρ_i, ρ_e together with $B_{\rho_i}(z) \subset \tilde{E}$. $\square$

Remark 11.11 (Why the covering vanished). The bookkeeping difficulty of Remark 11.7 was: many fjords, each charged to the global δ_{iso} without double counting. The coarea identity (25) does the bookkeeping in one stroke — $N(\rho)$ counts *all* crossings at radius ρ simultaneously, so $\int N d\rho$ is automatically additive over fjords with no risk of double counting, and the lone Cauchy–Schwarz of Step 2 converts it into the global deficit $P - P_r \leq \text{Exc} + Ca$. No pointwise upper density and no covering are used; the only ingredients are the deficit D_I (through Exc and, via FMP, through a), connectedness of $\tilde{E}$, and the no-hole property (Proposition 8.1). Notably D_H is not even needed for the trapping.

11.7 The regularity-free annulus

Theorem 11.12 (C^2 -free trapping). *Let $\delta_T \leq \delta_\star$ and let $\tilde{E}$ (its main component, cf. Remark 9.5) satisfy the two-sided density (15) with the absolute constant c_0 of Propositions 11.5–11.6 and*

$$r_* := \max\{r_*^{\text{int}}, r_*^{\text{ext}}\} \leq C \delta_T^{1/4}.$$

Then

$$B_{\hat{r}-\eta}(z) \subset \tilde{E} \subset B_{\hat{r}+\eta}(z), \quad \eta \leq C \delta_T^{1/4}, \quad (27)$$

with C absolute (depending only on c_0 and the universal constants of the deficit identities), and in particular not on the C^2 geometry of $\tilde{E}$.

Proof. Apply Lemma 11.1 with $A = a \leq C\sqrt{\delta_T}$ (Corollary 7.2) and $r_* \leq C\delta_T^{1/4}$: then $\eta = \max\{2r_*, 2\sqrt{a/(\pi c_0)}\} \leq C \max\{\delta_T^{1/4}, (\sqrt{\delta_T})^{1/2}\} = C\delta_T^{1/4}$. The density hypothesis (15) is Propositions 11.5 (lower, $r \geq r_*^{\text{int}} \leq C\delta_T$) and 11.6 (upper, $r \geq r_*^{\text{ext}} \leq C\delta_T^{1/4}$), valid for $r \in [r_*, r_0]$; the constants are absolute by those propositions. $\square$

Remark 11.13 (What this achieves, and the loop back to D_H, D_I). Theorem 11.12 replaces the C^2 -dependent constant of Theorem 9.2 by an *absolute* one: the only inputs are the two-sided density and the L^1 -closeness, and *both reduce to the deficits* — the lower density to D_H (the weighted Serrin defect $W \leq \frac{C_0}{4\pi}\delta_T$, Proposition 11.5), the upper density and the L^1 scale to D_I (the isoperimetric defect, FMP, and the fjord perimeter cost, Propositions 11.6 and Corollary 7.2). The annulus width $\eta \leq C\delta_T^{1/4}$ is governed by the exterior (fjord) scale $r_*^{\text{ext}} \sim \delta_{\text{iso}}^{1/2}$ and the L^1 scale $\sqrt{a} \sim \delta_T^{1/4}$. With the annulus now in hand *without* regularity, one may feed it back into the level-set identity: the level sets of u inside the annulus inherit the trapping, and a second application of $\int (D_H + D_I) = 4\pi\delta_T$ on the trapped configuration is the intended mechanism for improving the exponent — the program’s next step, not carried out here. The multi-fjord covering bookkeeping that was the lone residual is *removed* by Theorem 11.10: the angular-perimeter coarea identity sums all fjords at once, so the two-sided C^2 -free annulus $\rho_e - \rho_i \leq C\delta_T^{1/4}$ holds with absolute constants, conditioned only on connectedness of the main component (which carries area $\hat{\mu} - O(\delta_T)$, Remark 9.5).

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